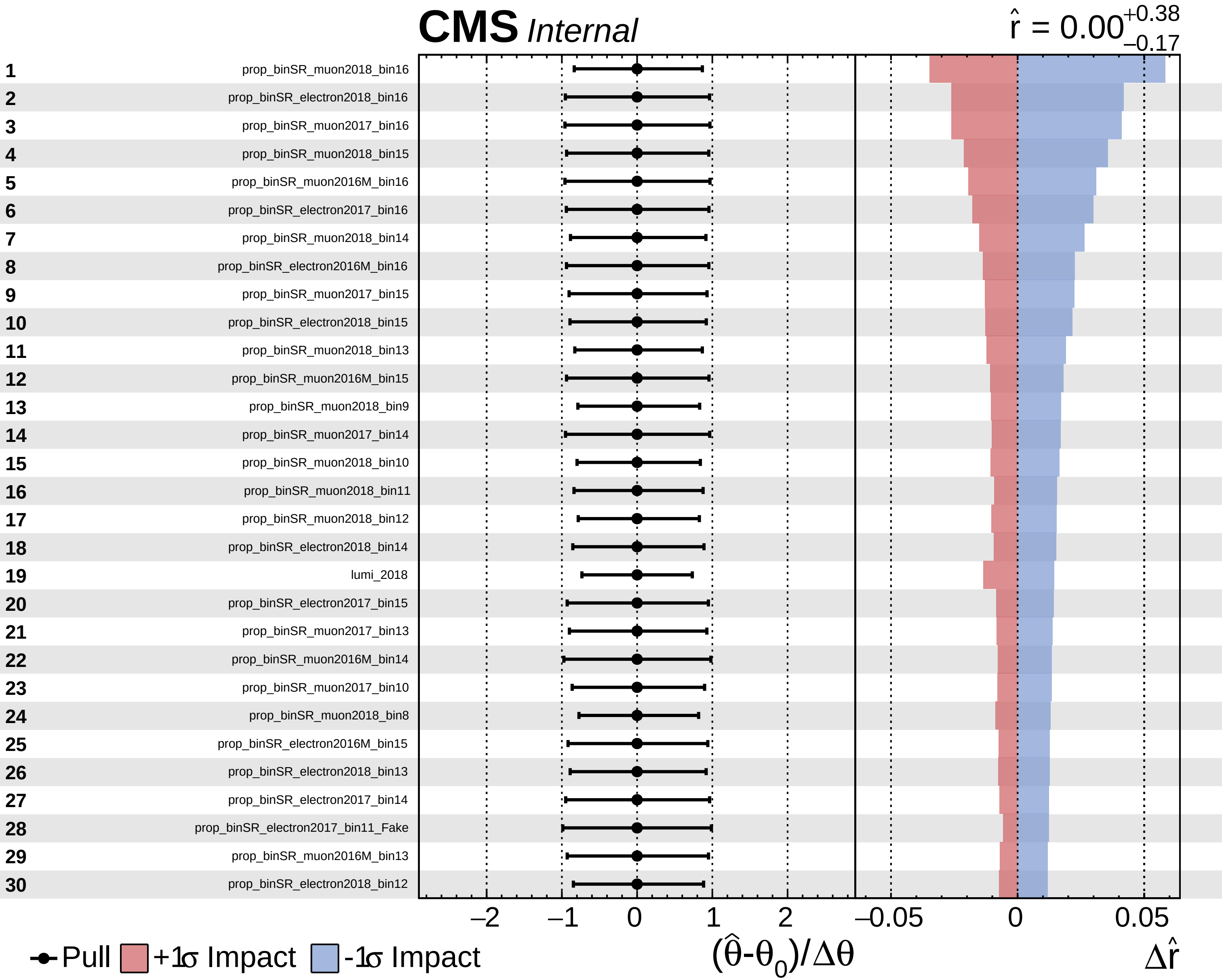
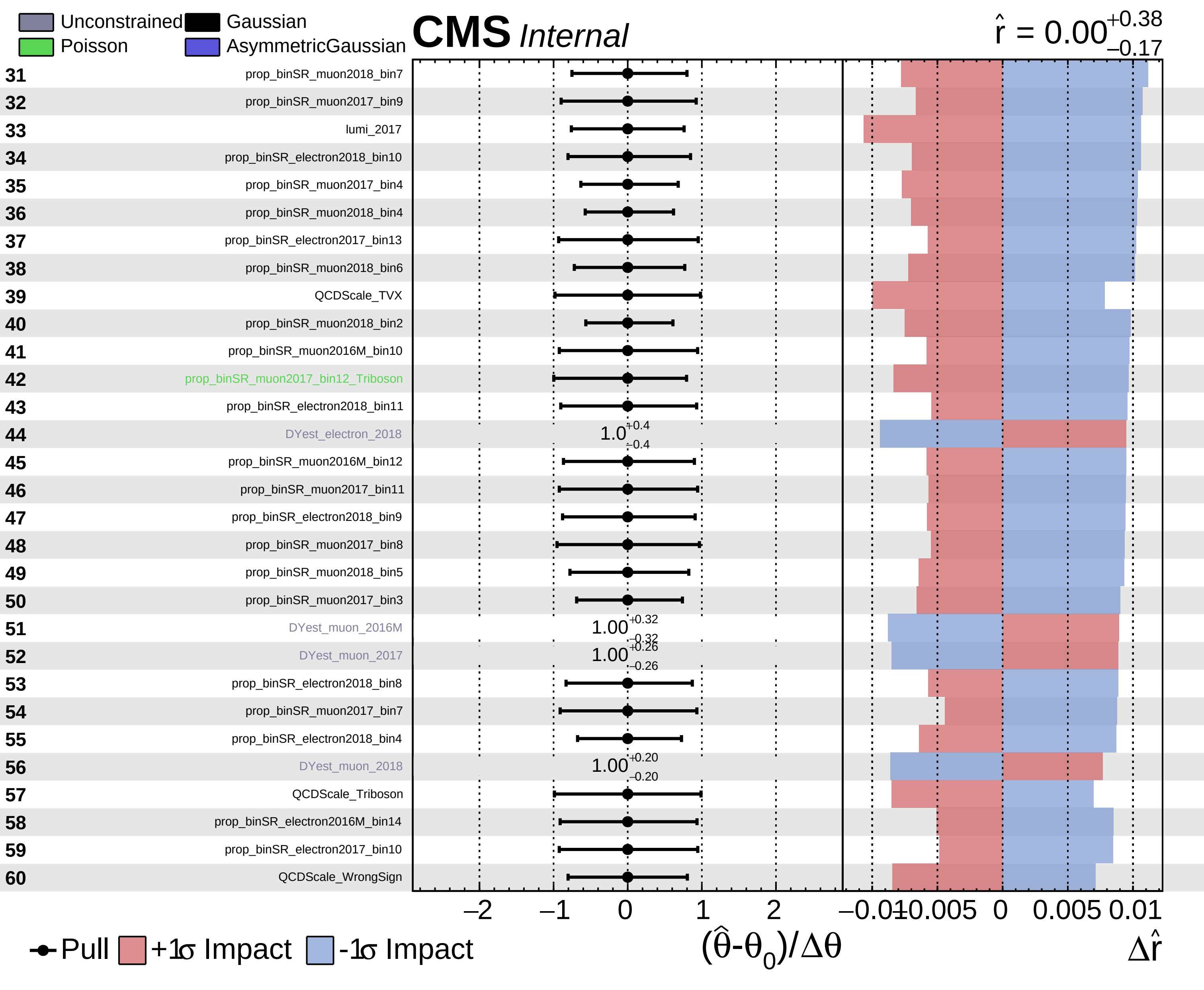
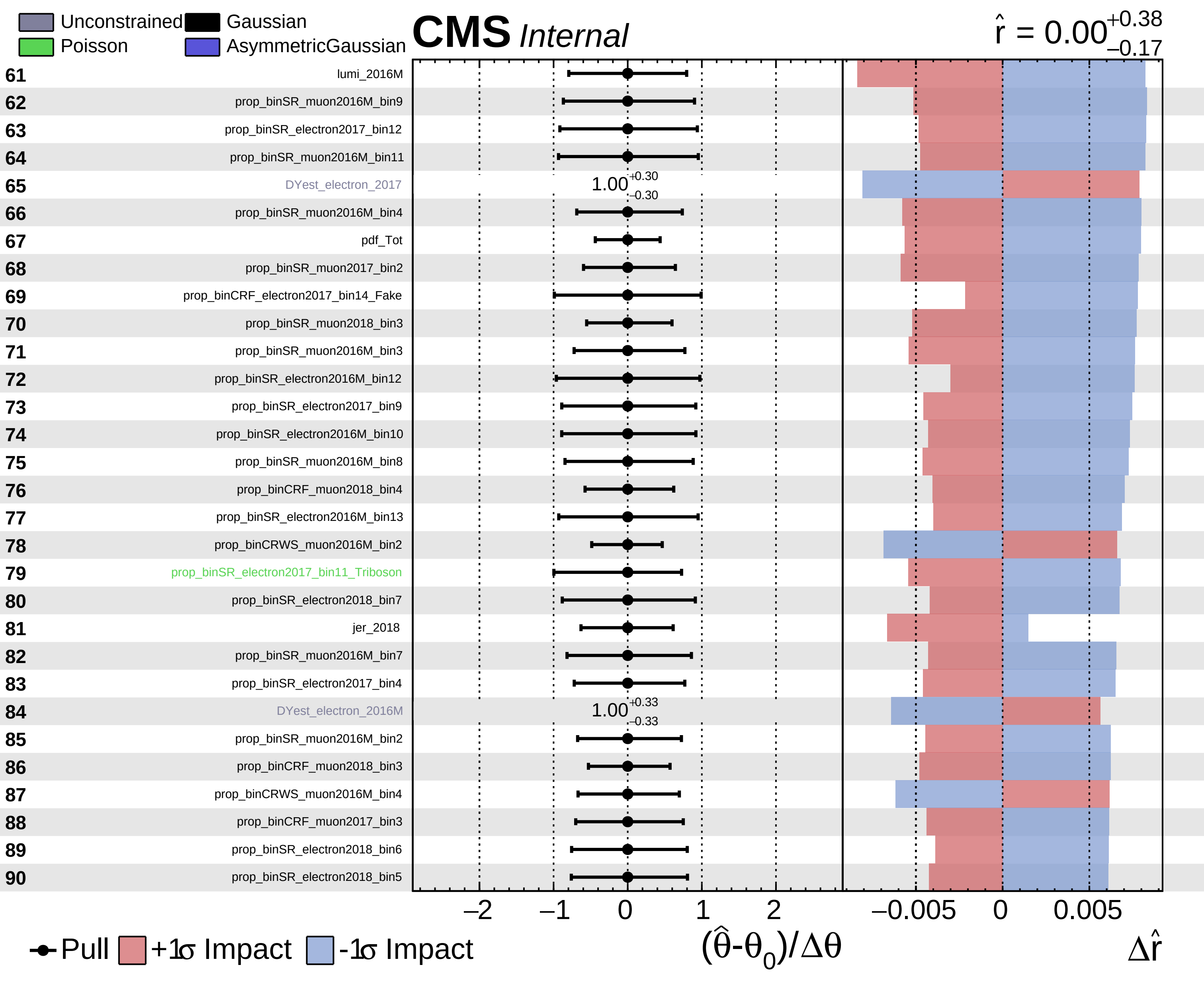


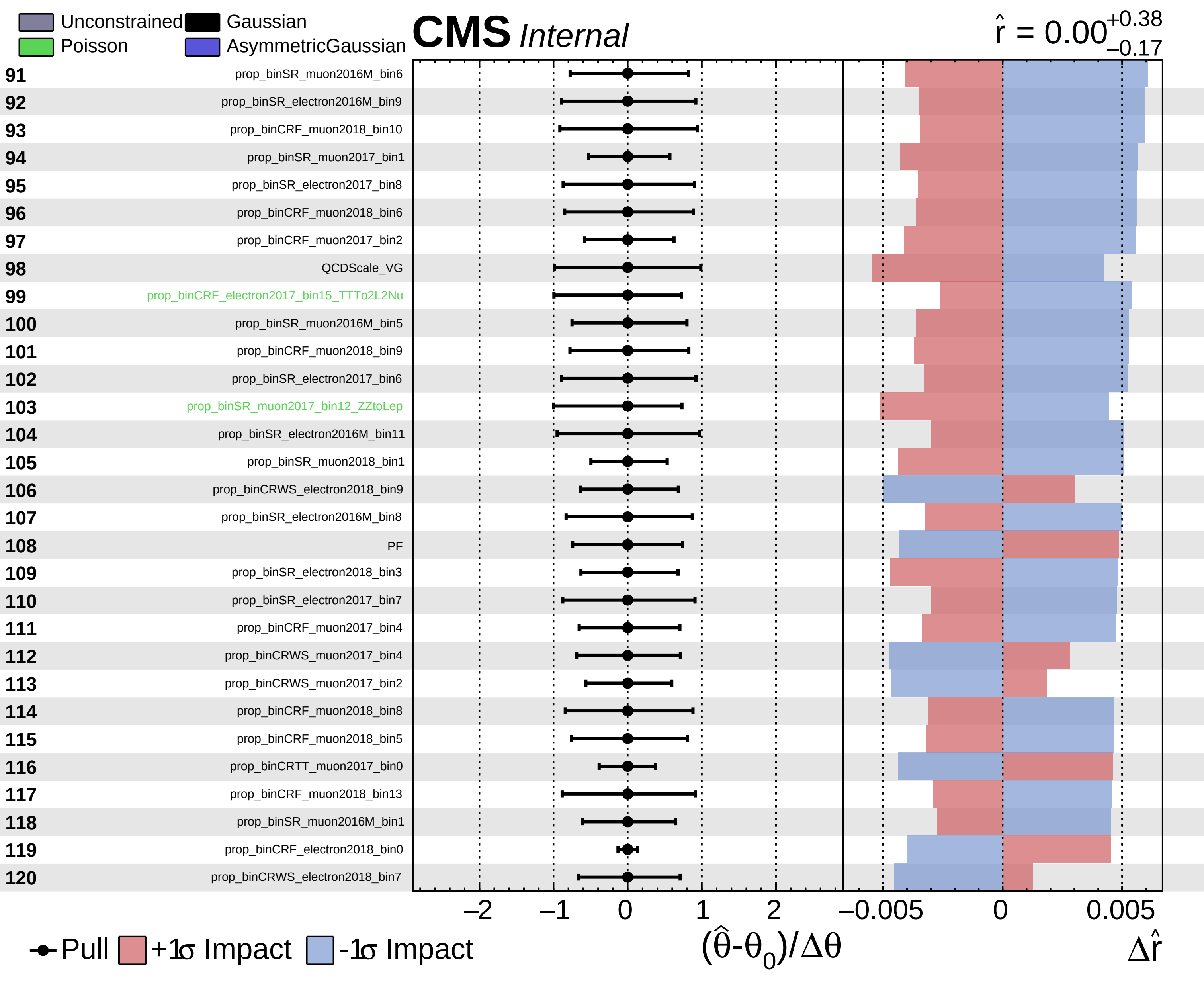
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 prop_binSR_electron2016M_bin16
9 prop_binSR_muon2017_bin15
10 prop_binSR_electron2018_bin15
11 prop_binSR_muon2018_bin13
12 prop_binSR_muon2016M_bin15
13 prop_binSR_muon2018_bin9
14 prop_binSR_muon2017_bin14
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2018_bin11
17 prop_binSR_muon2018_bin12
18 prop_binSR_electron2018_bin14
19 lumi_2018
20 prop_binSR_electron2017_bin15
21 prop_binSR_muon2017_bin13
22 prop_binSR_muon2016M_bin14
23 prop_binSR_muon2017_bin10
24 prop_binSR_muon2018_bin8
25 prop_binSR_electron2016M_bin15
26 prop_binSR_electron2018_bin13
27 prop_binSR_electron2017_bin14
28 prop_binSR_electron2017_bin11_Fake
29 prop_binSR_muon2016M_bin13
30 prop_binSR_electron2018_bin12
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

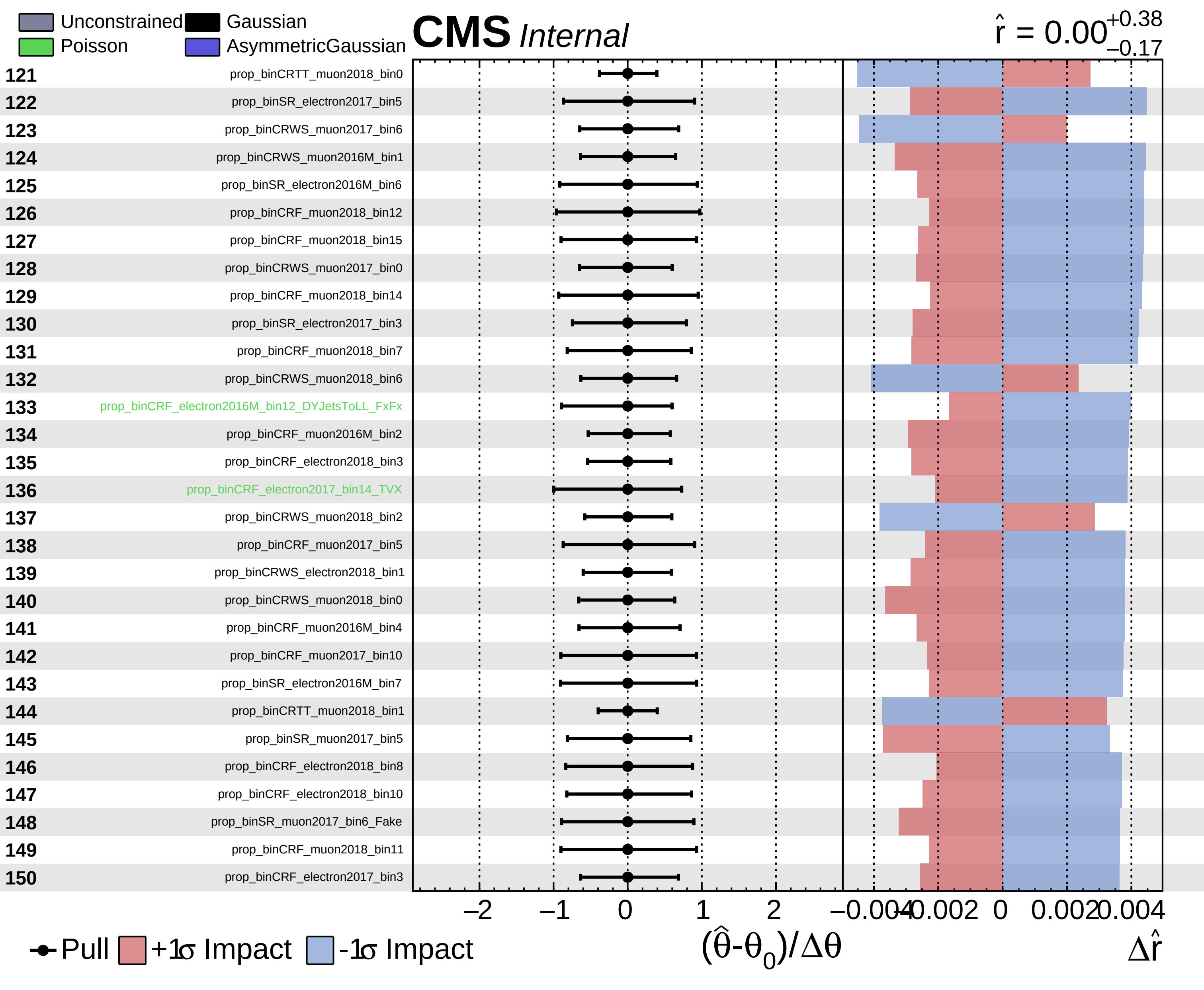


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.17
31 prop_binSR_muon2018_bin7
32 prop_binSR_muon2017_bin9
33 lumi_2017
34 prop_binSR_electron2018_bin10
35 prop_binSR_muon2017_bin4
36 prop_binSR_muon2018_bin4
37 prop_binSR_electron2017_bin13
38 prop_binSR_muon2018_bin6
39 QCDScale_TVX
40 prop_binSR_muon2018_bin2
41 prop_binSR_muon2016M_bin10
42 prop_binSR_muon2017_bin12_Triboson
43 prop_binSR_electron2018_bin11
44 DYest_electron_2018
1.0 +0.4 −0.4
45 prop_binSR_muon2016M_bin12
46 prop_binSR_muon2017_bin11
47 prop_binSR_electron2018_bin9
48 prop_binSR_muon2017_bin8
49 prop_binSR_muon2018_bin5
50 prop_binSR_muon2017_bin3
51 DYest_muon_2016M
1.00 +0.32 −0.32
52 DYest_muon_2017
1.00 +0.26 −0.26
53 prop_binSR_electron2018_bin8
54 prop_binSR_muon2017_bin7
55 prop_binSR_electron2018_bin4
56 DYest_muon_2018
1.00 +0.20 −0.20
57 QCDScale_Triboson
58 prop_binSR_electron2016M_bin14
59 prop_binSR_electron2017_bin10
60 QCDScale_WrongSign
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

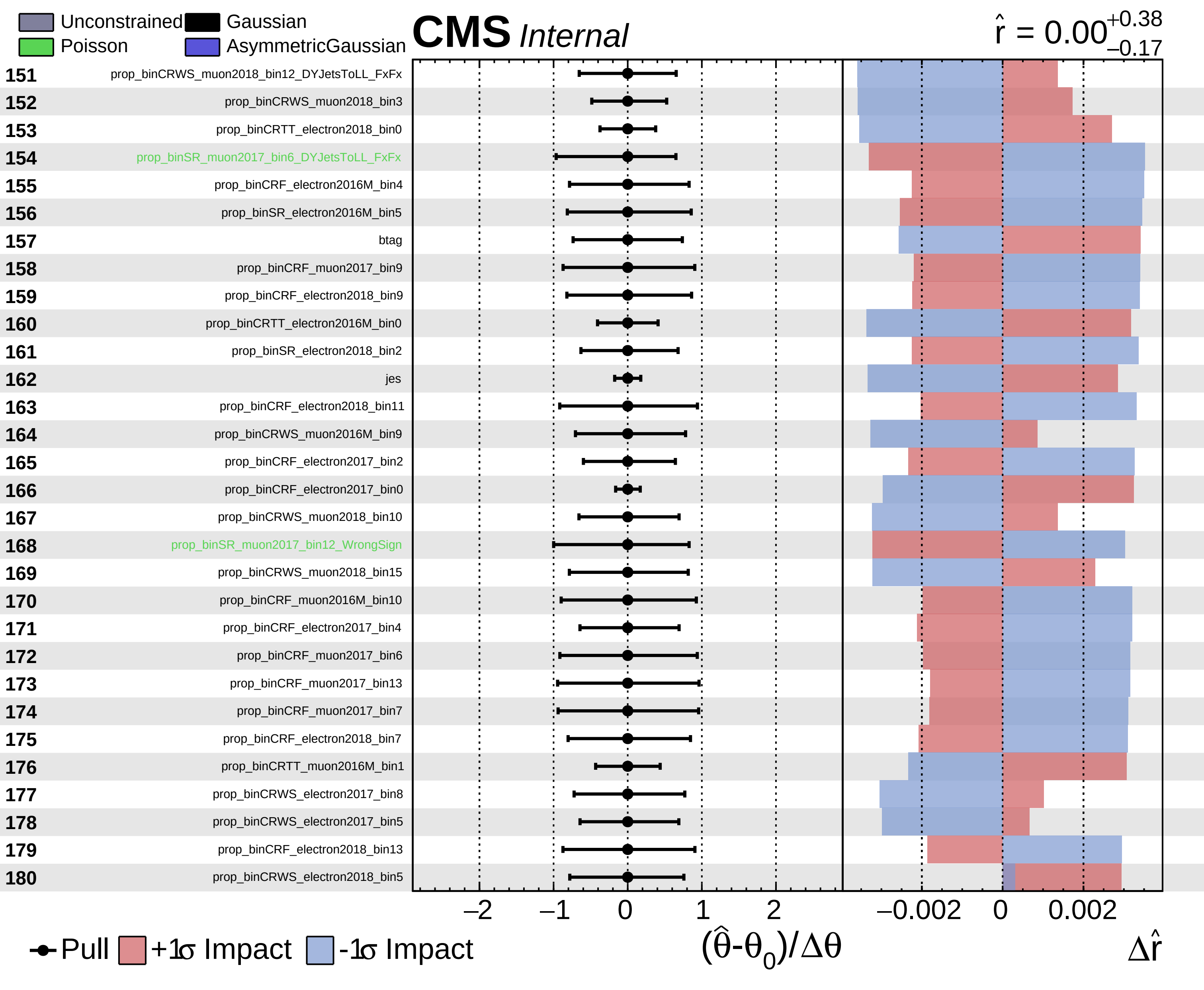


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.17
61 lumi_2016M
62 prop_binSR_muon2016M_bin9
63 prop_binSR_electron2017_bin12
64 prop_binSR_muon2016M_bin11
65 DYest_electron_2017
1.00 +0.30 −0.30
66 prop_binSR_muon2016M_bin4
67 pdf_Tot
68 prop_binSR_muon2017_bin2
69 prop_binCRF_electron2017_bin14_Fake
70 prop_binSR_muon2018_bin3
71 prop_binSR_muon2016M_bin3
72 prop_binSR_electron2016M_bin12
73 prop_binSR_electron2017_bin9
74 prop_binSR_electron2016M_bin10
75 prop_binSR_muon2016M_bin8
76 prop_binCRF_muon2018_bin4
77 prop_binSR_electron2016M_bin13
78 prop_binCRWS_muon2016M_bin2
79 prop_binSR_electron2017_bin11_Triboson
80 prop_binSR_electron2018_bin7
81 jer_2018
82 prop_binSR_muon2016M_bin7
83 prop_binSR_electron2017_bin4
84 DYest_electron_2016M
1.00 +0.33 −0.33
85 prop_binSR_muon2016M_bin2
86 prop_binCRF_muon2018_bin3
87 prop_binCRWS_muon2016M_bin4
88 prop_binCRF_muon2017_bin3
89 prop_binSR_electron2018_bin6
90 prop_binSR_electron2018_bin5
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

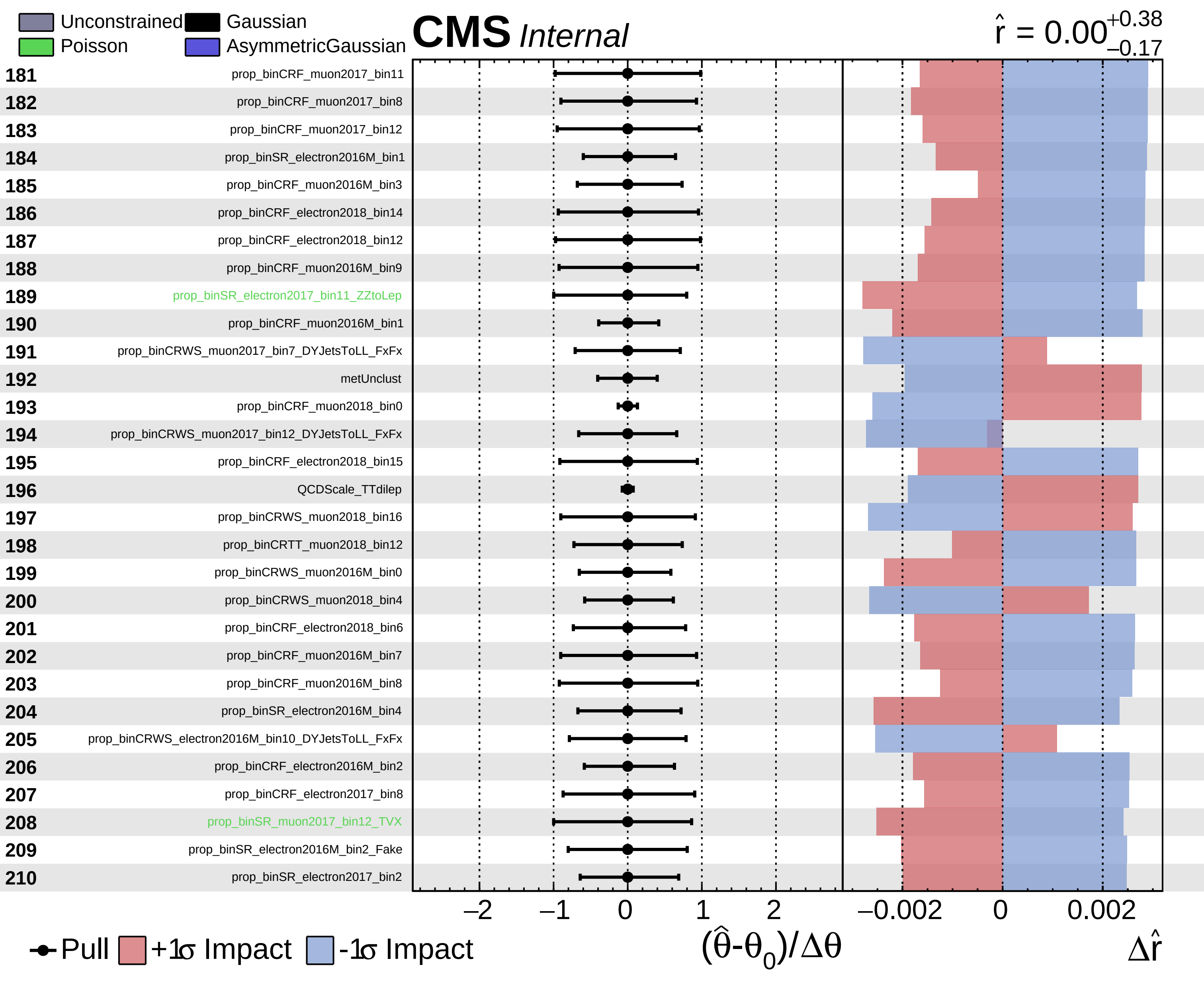

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.17
91 prop_binSR_muon2016M_bin6
92 prop_binSR_electron2016M_bin9
93 prop_binCRF_muon2018_bin10
94 prop_binSR_muon2017_bin1
95 prop_binSR_electron2017_bin8
96 prop_binCRF_muon2018_bin6
97 prop_binCRF_muon2017_bin2
98 QCDScale_VG
99 prop_binCRF_electron2017_bin15_TTTo2L2Nu
100 prop_binSR_muon2016M_bin5
101 prop_binCRF_muon2018_bin9
102 prop_binSR_electron2017_bin6
103 prop_binSR_muon2017_bin12_ZZtoLep
104 prop_binSR_electron2016M_bin11
105 prop_binSR_muon2018_bin1
106 prop_binCRWS_electron2018_bin9
107 prop_binSR_electron2016M_bin8
108 PF
109 prop_binSR_electron2018_bin3
110 prop_binSR_electron2017_bin7
111 prop_binCRF_muon2017_bin4
112 prop_binCRWS_muon2017_bin4
113 prop_binCRWS_muon2017_bin2
114 prop_binCRF_muon2018_bin8
115 prop_binCRF_muon2018_bin5
116 prop_binCRTT_muon2017_bin0
117 prop_binCRF_muon2018_bin13
118 prop_binSR_muon2016M_bin1
119 prop_binCRF_electron2018_bin0
120 prop_binCRWS_electron2018_bin7
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



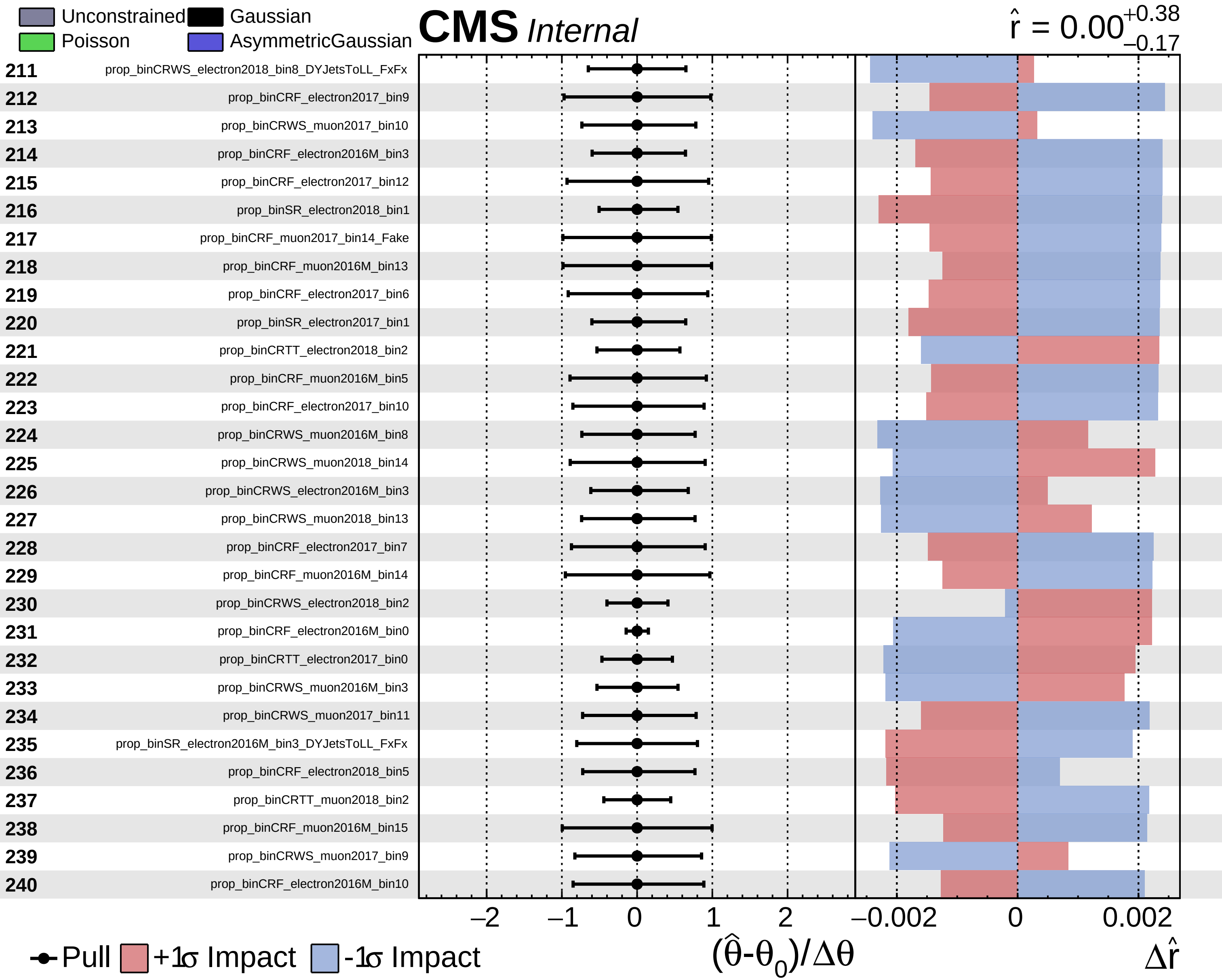
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
121 prop_binCRTT_muon2018_bin0
122 prop_binSR_electron2017_bin5
123 prop_binCRWS_muon2017_bin6
124 prop_binCRWS_muon2016M_bin1
125 prop_binSR_electron2016M_bin6
126 prop_binCRF_muon2018_bin12
127 prop_binCRF_muon2018_bin15
128 prop_binCRWS_muon2017_bin0
129 prop_binCRF_muon2018_bin14
130 prop_binSR_electron2017_bin3
131 prop_binCRF_muon2018_bin7
132 prop_binCRWS_muon2018_bin6
133 prop_binCRF_electron2016M_bin12_DYJetsToLL_FxFx
134 prop_binCRF_muon2016M_bin2
135 prop_binCRF_electron2018_bin3
136 prop_binCRF_electron2017_bin14_TVX
137 prop_binCRWS_muon2018_bin2
138 prop_binCRF_muon2017_bin5
139 prop_binCRWS_electron2018_bin1
140 prop_binCRWS_muon2018_bin0
141 prop_binCRF_muon2016M_bin4
142 prop_binCRF_muon2017_bin10
143 prop_binSR_electron2016M_bin7
144 prop_binCRTT_muon2018_bin1
145 prop_binSR_muon2017_bin5
146 prop_binCRF_electron2018_bin8
147 prop_binCRF_electron2018_bin10
148 prop_binSR_muon2017_bin6_Fake
149 prop_binCRF_muon2018_bin11
150 prop_binCRF_electron2017_bin3
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



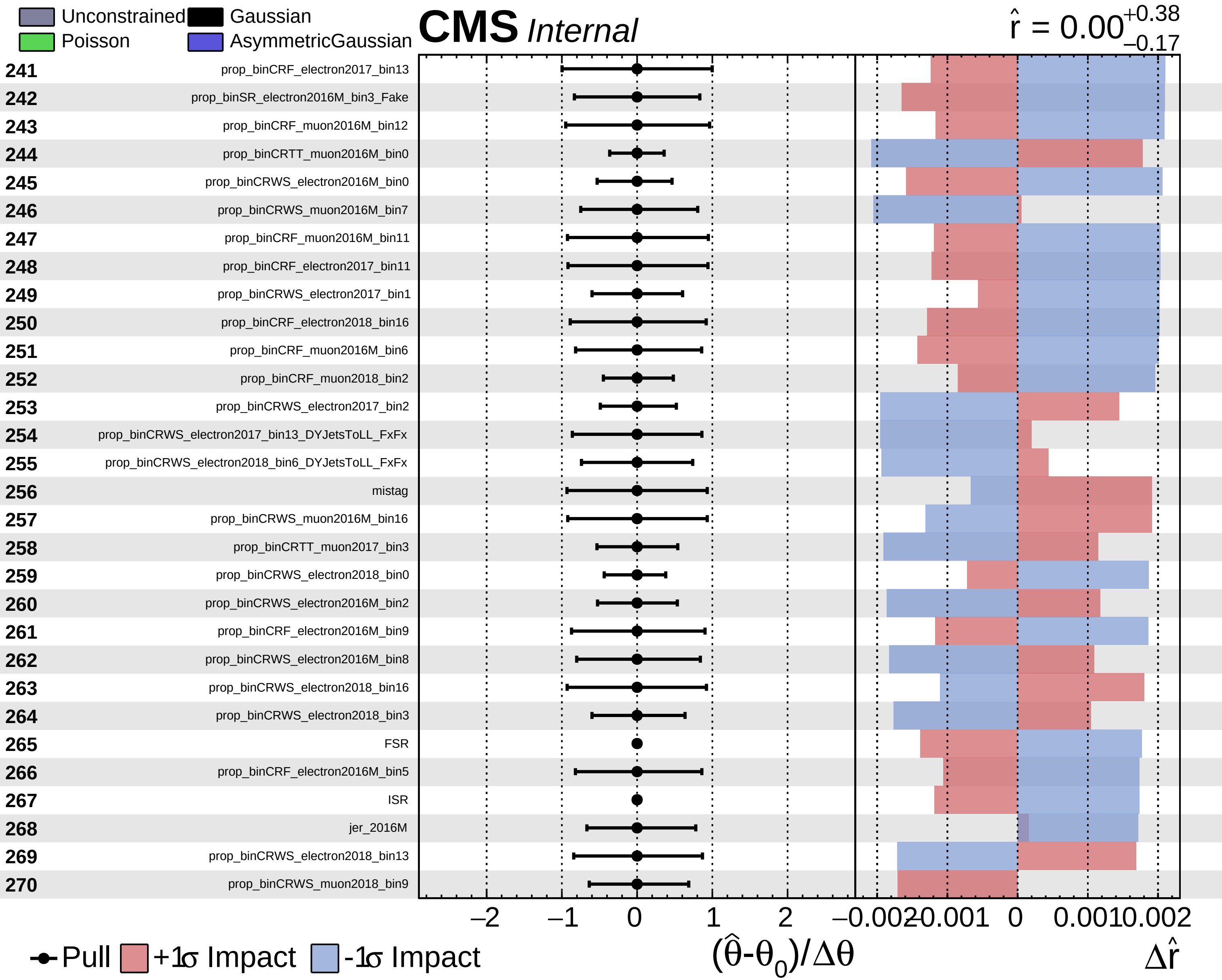
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
151 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
152 prop_binCRWS_muon2018_bin3
153 prop_binCRTT_electron2018_bin0
154 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
155 prop_binCRF_electron2016M_bin4
156 prop_binSR_electron2016M_bin5
157 btag
158 prop_binCRF_muon2017_bin9
159 prop_binCRF_electron2018_bin9
160 prop_binCRTT_electron2016M_bin0
161 prop_binSR_electron2018_bin2
162 jes
163 prop_binCRF_electron2018_bin11
164 prop_binCRWS_muon2016M_bin9
165 prop_binCRF_electron2017_bin2
166 prop_binCRF_electron2017_bin0
167 prop_binCRWS_muon2018_bin10
168 prop_binSR_muon2017_bin12_WrongSign
169 prop_binCRWS_muon2018_bin15
170 prop_binCRF_muon2016M_bin10
171 prop_binCRF_electron2017_bin4
172 prop_binCRF_muon2017_bin6
173 prop_binCRF_muon2017_bin13
174 prop_binCRF_muon2017_bin7
175 prop_binCRF_electron2018_bin7
176 prop_binCRTT_muon2016M_bin1
177 prop_binCRWS_electron2017_bin8
178 prop_binCRWS_electron2017_bin5
179 prop_binCRF_electron2018_bin13
180 prop_binCRWS_electron2018_bin5
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



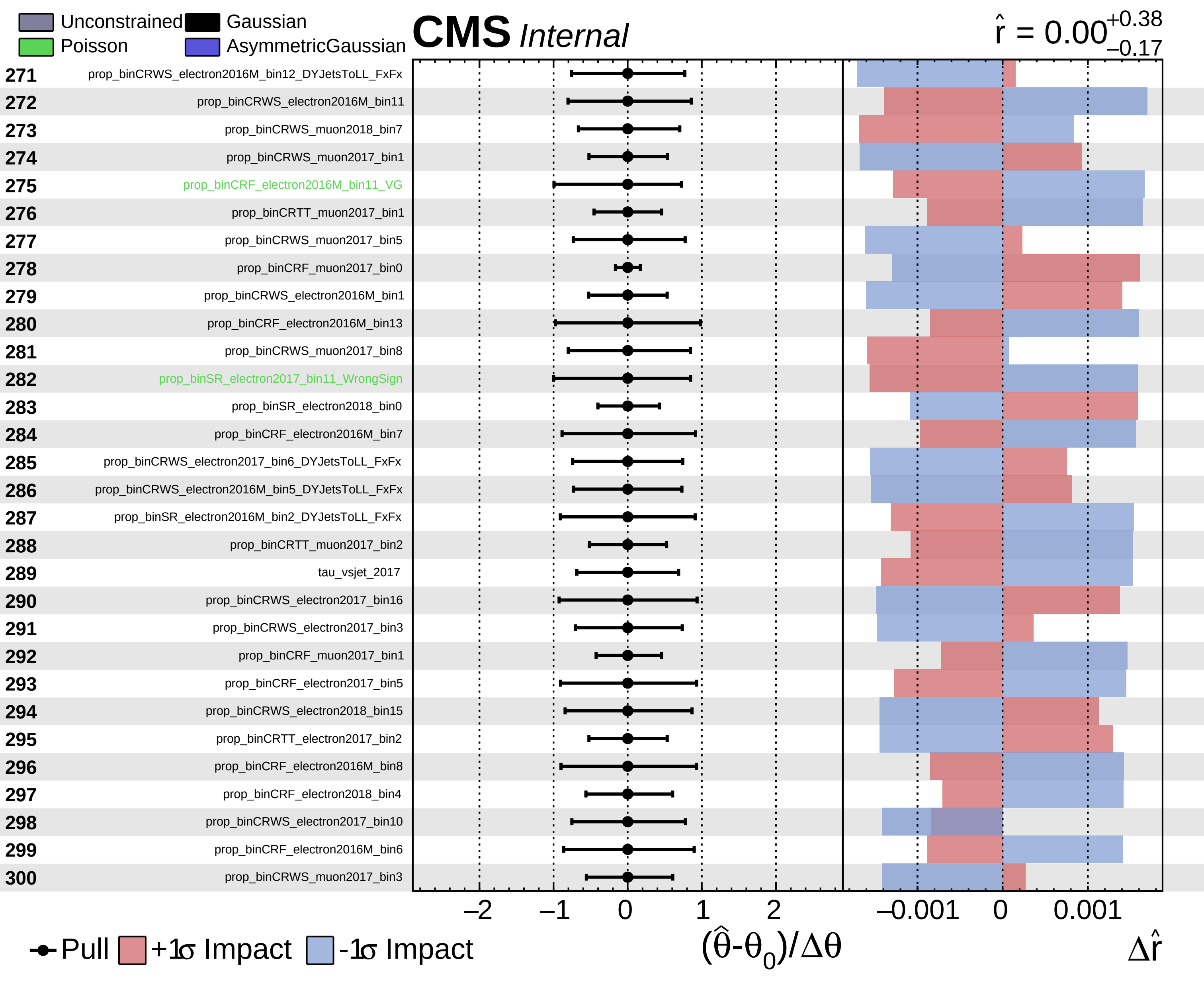
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
181 prop_binCRF_muon2017_bin11
182 prop_binCRF_muon2017_bin8
183 prop_binCRF_muon2017_bin12
184 prop_binSR_electron2016M_bin1
185 prop_binCRF_muon2016M_bin3
186 prop_binCRF_electron2018_bin14
187 prop_binCRF_electron2018_bin12
188 prop_binCRF_muon2016M_bin9
189 prop_binSR_electron2017_bin11_ZZtoLep
190 prop_binCRF_muon2016M_bin1
191 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
192 metUnclust
193 prop_binCRF_muon2018_bin0
194 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
195 prop_binCRF_electron2018_bin15
196 QCDScale_TTdilep
197 prop_binCRWS_muon2018_bin16
198 prop_binCRTT_muon2018_bin12
199 prop_binCRWS_muon2016M_bin0
200 prop_binCRWS_muon2018_bin4
201 prop_binCRF_electron2018_bin6
202 prop_binCRF_muon2016M_bin7
203 prop_binCRF_muon2016M_bin8
204 prop_binSR_electron2016M_bin4
205 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
206 prop_binCRF_electron2016M_bin2
207 prop_binCRF_electron2017_bin8
208 prop_binSR_muon2017_bin12_TVX
209 prop_binSR_electron2016M_bin2_Fake
210 prop_binSR_electron2017_bin2
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



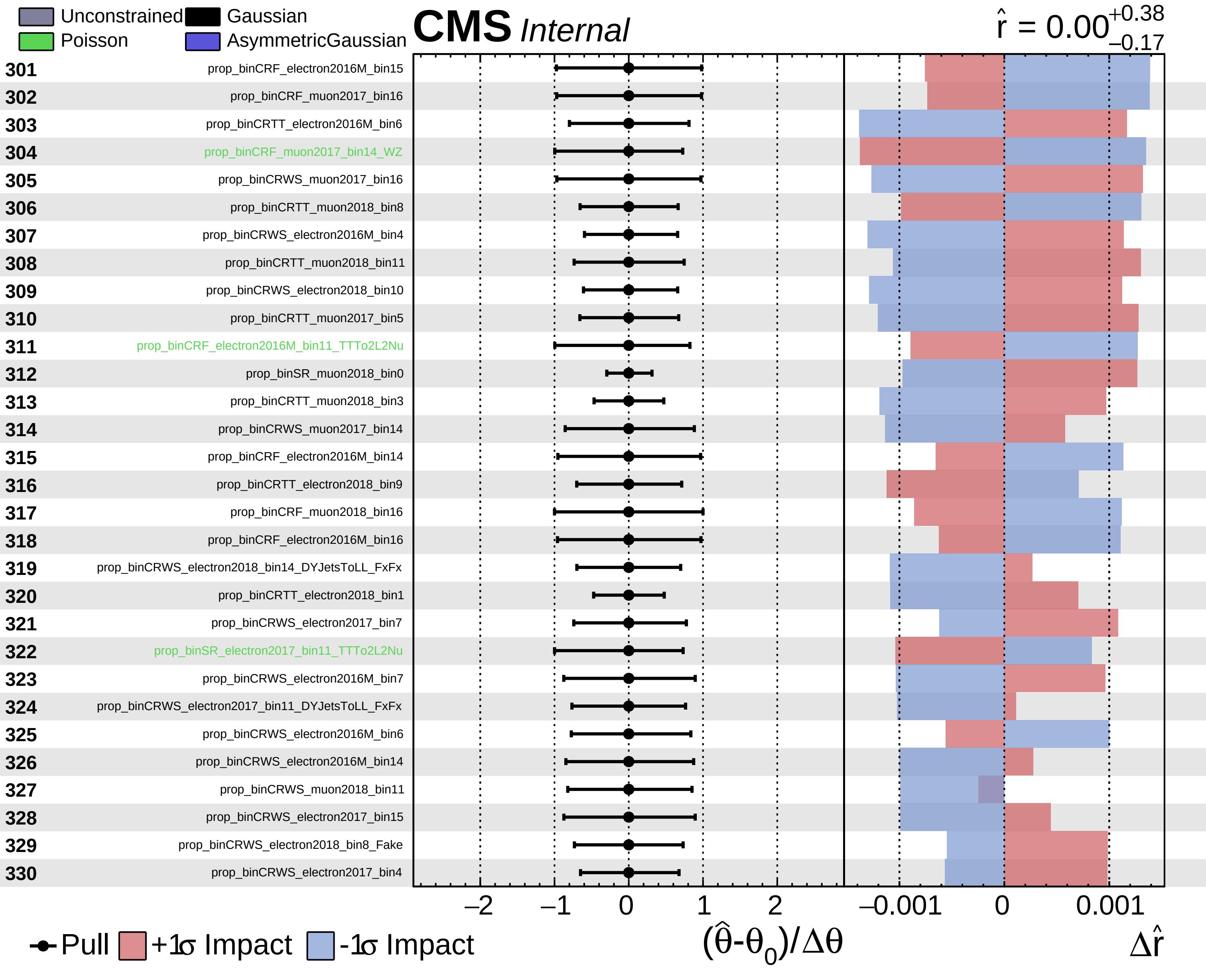
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
211 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
212 prop_binCRF_electron2017_bin9
213 prop_binCRWS_muon2017_bin10
214 prop_binCRF_electron2016M_bin3
215 prop_binCRF_electron2017_bin12
216 prop_binSR_electron2018_bin1
217 prop_binCRF_muon2017_bin14_Fake
218 prop_binCRF_muon2016M_bin13
219 prop_binCRF_electron2017_bin6
220 prop_binSR_electron2017_bin1
221 prop_binCRTT_electron2018_bin2
222 prop_binCRF_muon2016M_bin5
223 prop_binCRF_electron2017_bin10
224 prop_binCRWS_muon2016M_bin8
225 prop_binCRWS_muon2018_bin14
226 prop_binCRWS_electron2016M_bin3
227 prop_binCRWS_muon2018_bin13
228 prop_binCRF_electron2017_bin7
229 prop_binCRF_muon2016M_bin14
230 prop_binCRWS_electron2018_bin2
231 prop_binCRF_electron2016M_bin0
232 prop_binCRTT_electron2017_bin0
233 prop_binCRWS_muon2016M_bin3
234 prop_binCRWS_muon2017_bin11
235 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
236 prop_binCRF_electron2018_bin5
237 prop_binCRTT_muon2018_bin2
238 prop_binCRF_muon2016M_bin15
239 prop_binCRWS_muon2017_bin9
240 prop_binCRF_electron2016M_bin10
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



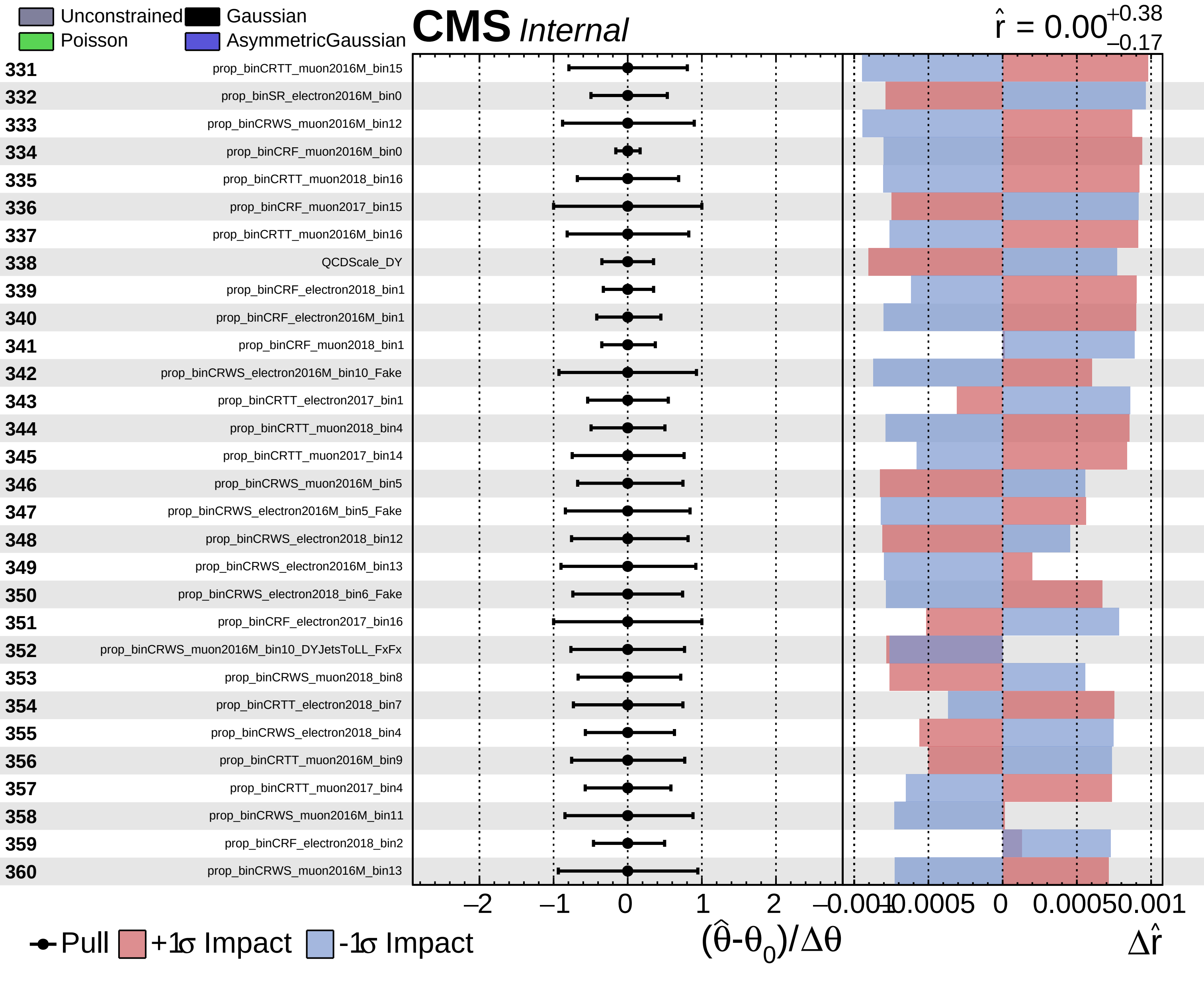
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
241 prop_binCRF_electron2017_bin13
242 prop_binSR_electron2016M_bin3_Fake
243 prop_binCRF_muon2016M_bin12
244 prop_binCRTT_muon2016M_bin0
245 prop_binCRWS_electron2016M_bin0
246 prop_binCRWS_muon2016M_bin7
247 prop_binCRF_muon2016M_bin11
248 prop_binCRF_electron2017_bin11
249 prop_binCRWS_electron2017_bin1
250 prop_binCRF_electron2018_bin16
251 prop_binCRF_muon2016M_bin6
252 prop_binCRF_muon2018_bin2
253 prop_binCRWS_electron2017_bin2
254 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
255 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
256 mistag
257 prop_binCRWS_muon2016M_bin16
258 prop_binCRTT_muon2017_bin3
259 prop_binCRWS_electron2018_bin0
260 prop_binCRWS_electron2016M_bin2
261 prop_binCRF_electron2016M_bin9
262 prop_binCRWS_electron2016M_bin8
263 prop_binCRWS_electron2018_bin16
264 prop_binCRWS_electron2018_bin3
265 FSR
266 prop_binCRF_electron2016M_bin5
267 ISR
268 jer_2016M
269 prop_binCRWS_electron2018_bin13
270 prop_binCRWS_muon2018_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



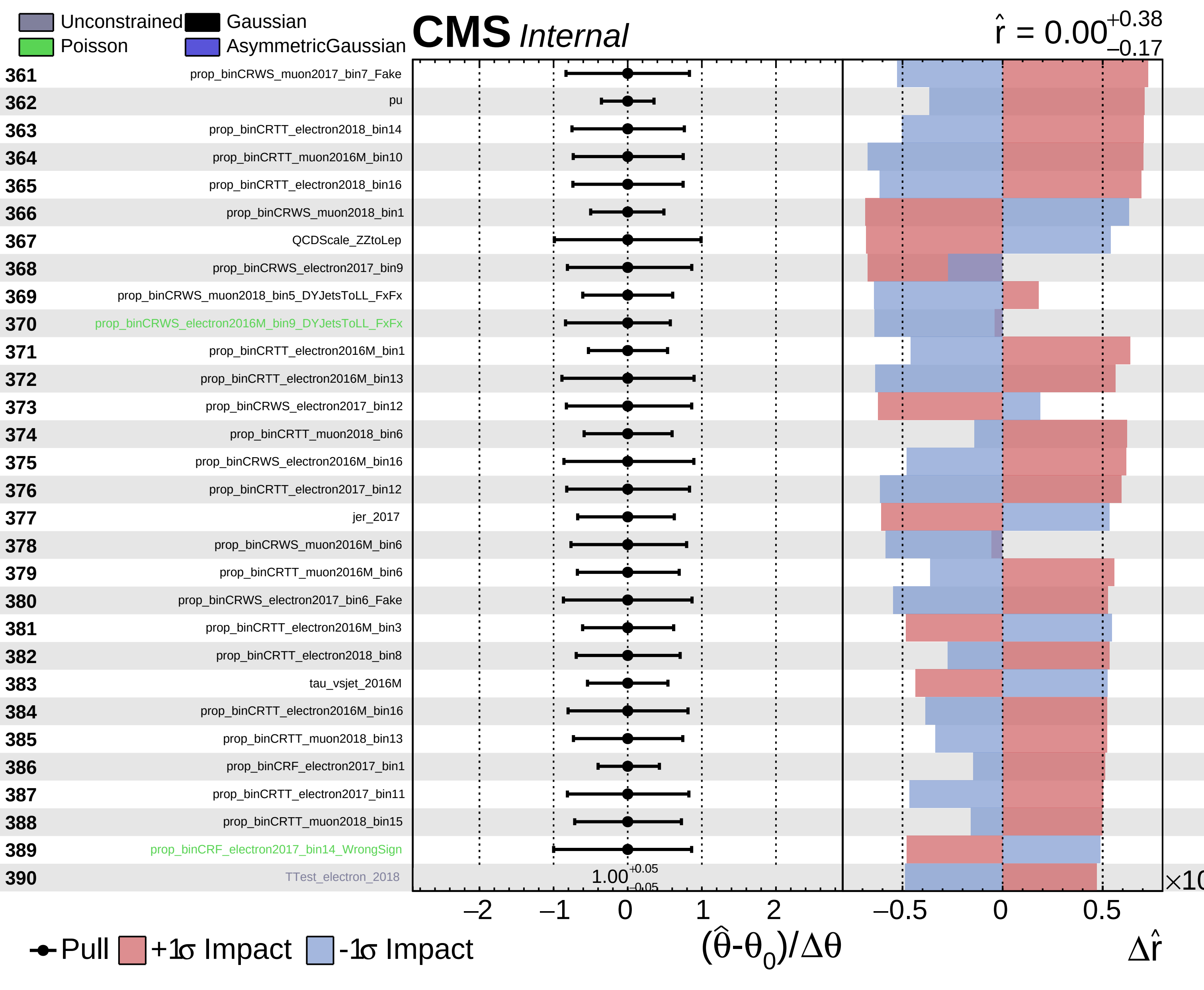
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
271 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
272 prop_binCRWS_electron2016M_bin11
273 prop_binCRWS_muon2018_bin7
274 prop_binCRWS_muon2017_bin1
275 prop_binCRF_electron2016M_bin11_VG
276 prop_binCRTT_muon2017_bin1
277 prop_binCRWS_muon2017_bin5
278 prop_binCRF_muon2017_bin0
279 prop_binCRWS_electron2016M_bin1
280 prop_binCRF_electron2016M_bin13
281 prop_binCRWS_muon2017_bin8
282 prop_binSR_electron2017_bin11_WrongSign
283 prop_binSR_electron2018_bin0
284 prop_binCRF_electron2016M_bin7
285 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
286 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
287 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
288 prop_binCRTT_muon2017_bin2
289 tau_vsjet_2017
290 prop_binCRWS_electron2017_bin16
291 prop_binCRWS_electron2017_bin3
292 prop_binCRF_muon2017_bin1
293 prop_binCRF_electron2017_bin5
294 prop_binCRWS_electron2018_bin15
295 prop_binCRTT_electron2017_bin2
296 prop_binCRF_electron2016M_bin8
297 prop_binCRF_electron2018_bin4
298 prop_binCRWS_electron2017_bin10
299 prop_binCRF_electron2016M_bin6
300 prop_binCRWS_muon2017_bin3
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



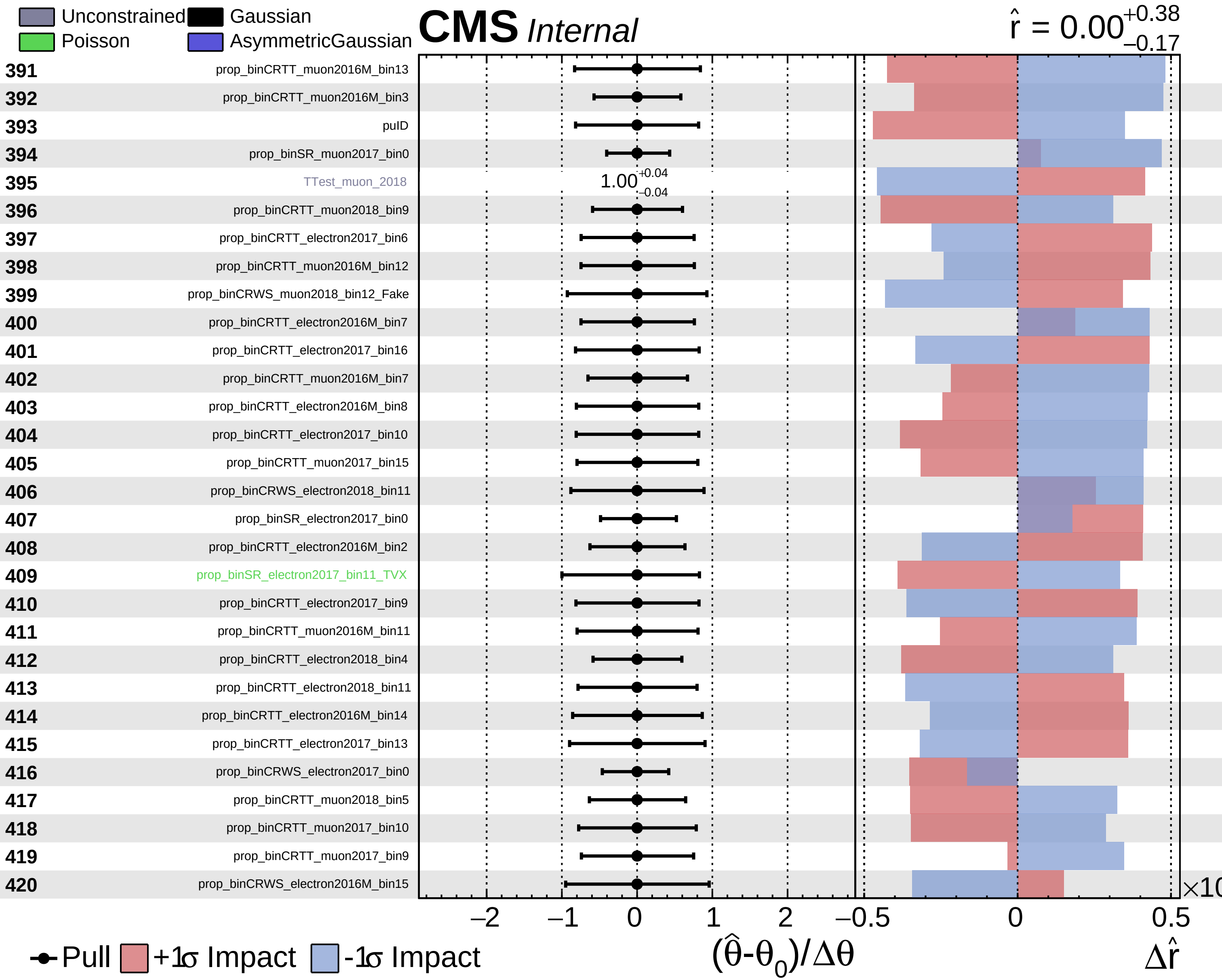
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
301 prop_binCRF_electron2016M_bin15
302 prop_binCRF_muon2017_bin16
303 prop_binCRTT_electron2016M_bin6
304 prop_binCRF_muon2017_bin14_WZ
305 prop_binCRWS_muon2017_bin16
306 prop_binCRTT_muon2018_bin8
307 prop_binCRWS_electron2016M_bin4
308 prop_binCRTT_muon2018_bin11
309 prop_binCRWS_electron2018_bin10
310 prop_binCRTT_muon2017_bin5
311 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
312 prop_binSR_muon2018_bin0
313 prop_binCRTT_muon2018_bin3
314 prop_binCRWS_muon2017_bin14
315 prop_binCRF_electron2016M_bin14
316 prop_binCRTT_electron2018_bin9
317 prop_binCRF_muon2018_bin16
318 prop_binCRF_electron2016M_bin16
319 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
320 prop_binCRTT_electron2018_bin1
321 prop_binCRWS_electron2017_bin7
322 prop_binSR_electron2017_bin11_TTTo2L2Nu
323 prop_binCRWS_electron2016M_bin7
324 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
325 prop_binCRWS_electron2016M_bin6
326 prop_binCRWS_electron2016M_bin14
327 prop_binCRWS_muon2018_bin11
328 prop_binCRWS_electron2017_bin15
329 prop_binCRWS_electron2018_bin8_Fake
330 prop_binCRWS_electron2017_bin4
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



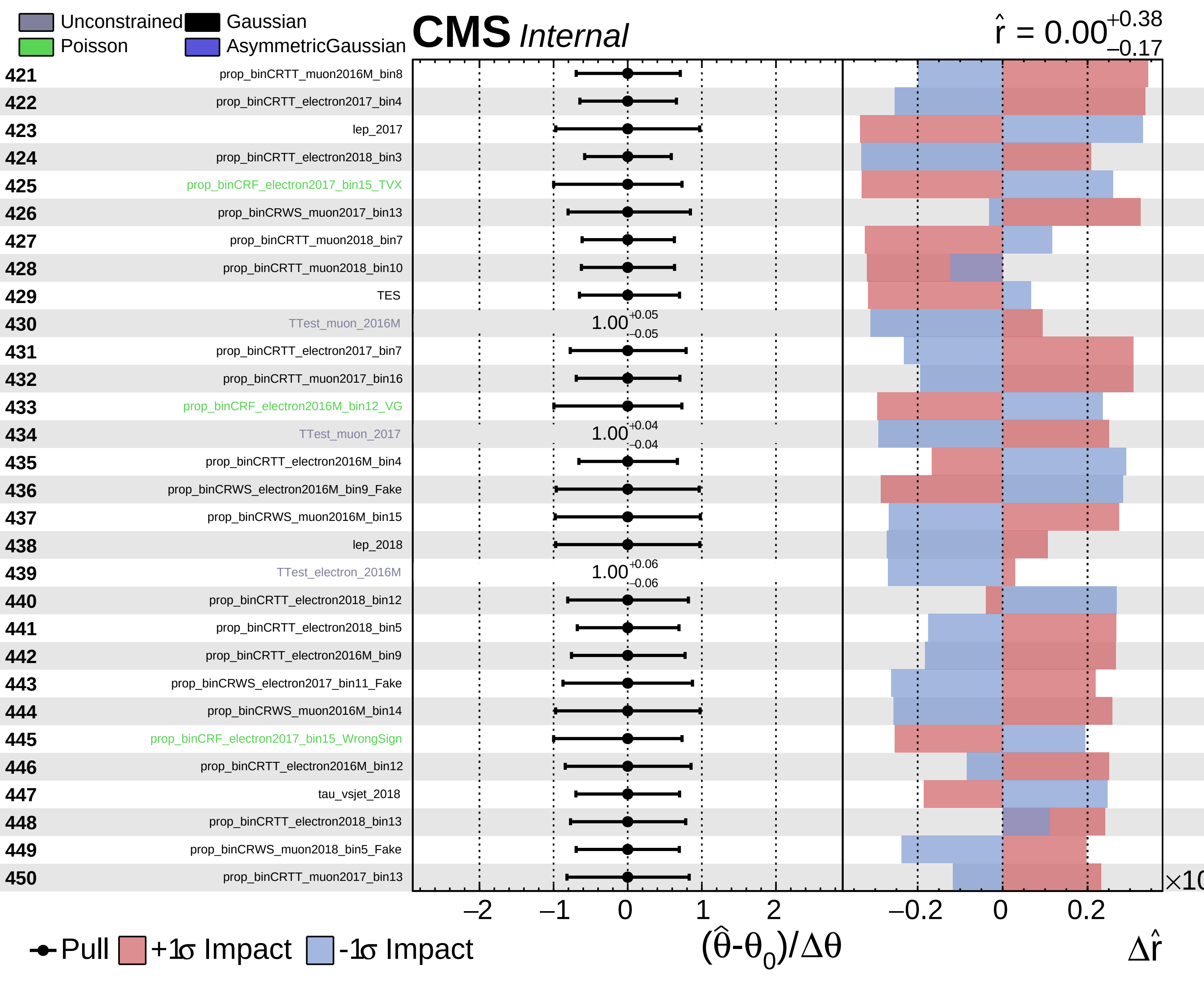
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
331 prop_binCRTT_muon2016M_bin15
332 prop_binSR_electron2016M_bin0
333 prop_binCRWS_muon2016M_bin12
334 prop_binCRF_muon2016M_bin0
335 prop_binCRTT_muon2018_bin16
336 prop_binCRF_muon2017_bin15
337 prop_binCRTT_muon2016M_bin16
338 QCDScale_DY
339 prop_binCRF_electron2018_bin1
340 prop_binCRF_electron2016M_bin1
341 prop_binCRF_muon2018_bin1
342 prop_binCRWS_electron2016M_bin10_Fake
343 prop_binCRTT_electron2017_bin1
344 prop_binCRTT_muon2018_bin4
345 prop_binCRTT_muon2017_bin14
346 prop_binCRWS_muon2016M_bin5
347 prop_binCRWS_electron2016M_bin5_Fake
348 prop_binCRWS_electron2018_bin12
349 prop_binCRWS_electron2016M_bin13
350 prop_binCRWS_electron2018_bin6_Fake
351 prop_binCRF_electron2017_bin16
352 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
353 prop_binCRWS_muon2018_bin8
354 prop_binCRTT_electron2018_bin7
355 prop_binCRWS_electron2018_bin4
356 prop_binCRTT_muon2016M_bin9
357 prop_binCRTT_muon2017_bin4
358 prop_binCRWS_muon2016M_bin11
359 prop_binCRF_electron2018_bin2
360 prop_binCRWS_muon2016M_bin13
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

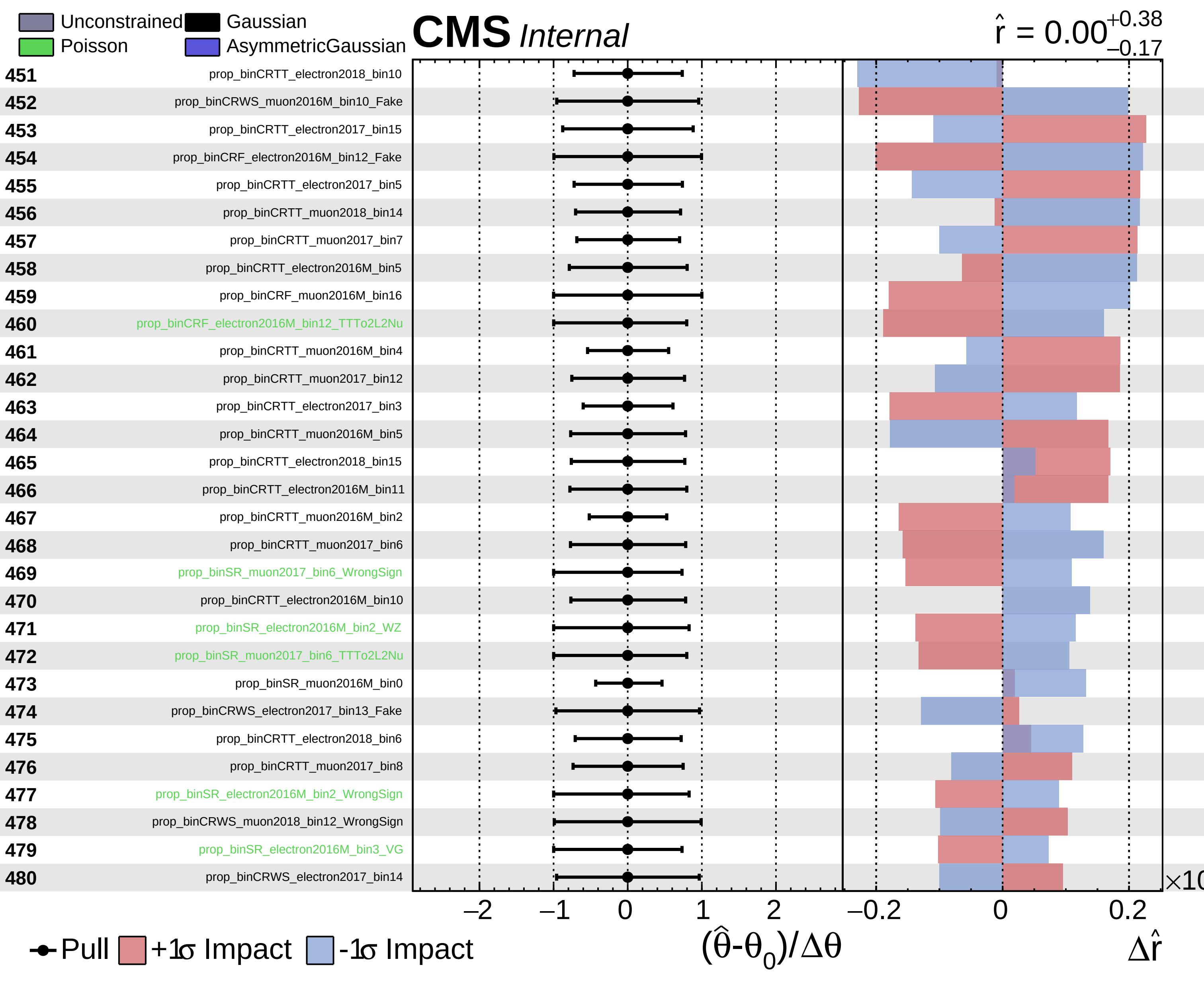


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
361 prop_binCRWS_muon2017_bin7_Fake
362 pu
363 prop_binCRTT_electron2018_bin14
364 prop_binCRTT_muon2016M_bin10
365 prop_binCRTT_electron2018_bin16
366 prop_binCRWS_muon2018_bin1
367 QCDScale_ZZtoLep
368 prop_binCRWS_electron2017_bin9
369 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
370 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
371 prop_binCRTT_electron2016M_bin1
372 prop_binCRTT_electron2016M_bin13
373 prop_binCRWS_electron2017_bin12
374 prop_binCRTT_muon2018_bin6
375 prop_binCRWS_electron2016M_bin16
376 prop_binCRTT_electron2017_bin12
377 jer_2017
378 prop_binCRWS_muon2016M_bin6
379 prop_binCRTT_muon2016M_bin6
380 prop_binCRWS_electron2017_bin6_Fake
381 prop_binCRTT_electron2016M_bin3
382 prop_binCRTT_electron2018_bin8
383 tau_vsjet_2016M
384 prop_binCRTT_electron2016M_bin16
385 prop_binCRTT_muon2018_bin13
386 prop_binCRF_electron2017_bin1
387 prop_binCRTT_electron2017_bin11
388 prop_binCRTT_muon2018_bin15
389 prop_binCRF_electron2017_bin14_WrongSign
390 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

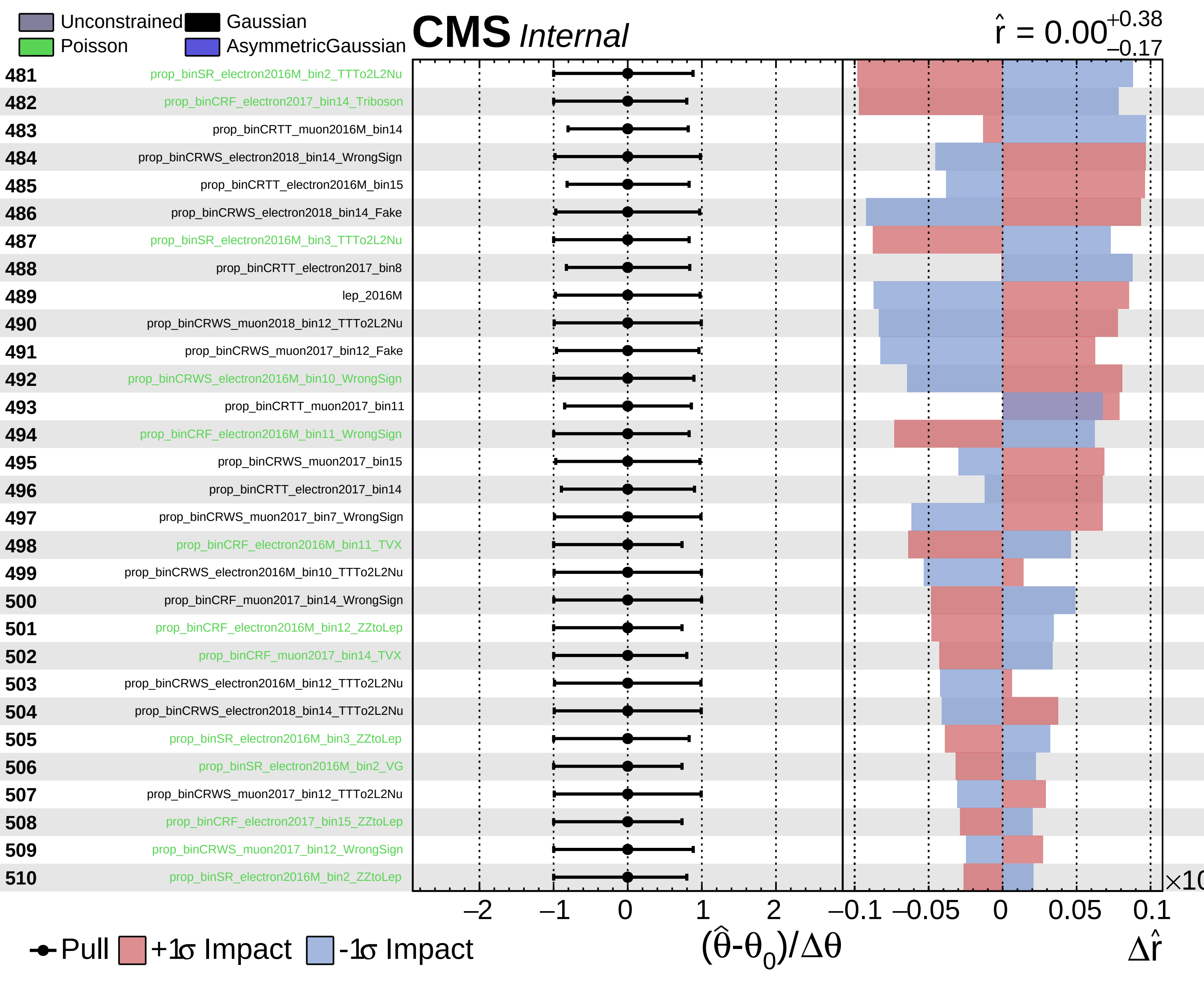


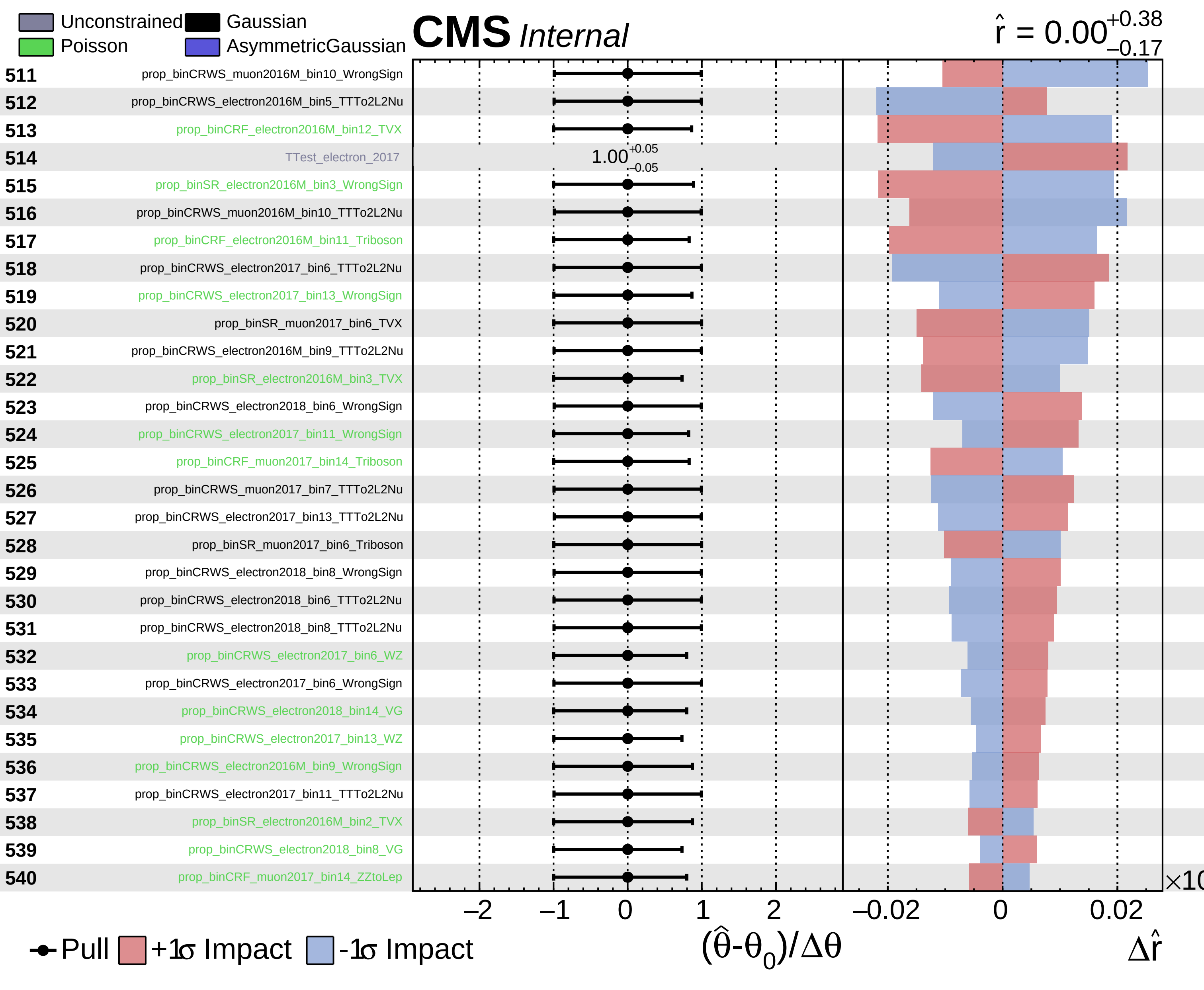
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
391 prop_binCRTT_muon2016M_bin13
392 prop_binCRTT_muon2016M_bin3
393 puID
394 prop_binSR_muon2017_bin0
395 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
396 prop_binCRTT_muon2018_bin9
397 prop_binCRTT_electron2017_bin6
398 prop_binCRTT_muon2016M_bin12
399 prop_binCRWS_muon2018_bin12_Fake
400 prop_binCRTT_electron2016M_bin7
401 prop_binCRTT_electron2017_bin16
402 prop_binCRTT_muon2016M_bin7
403 prop_binCRTT_electron2016M_bin8
404 prop_binCRTT_electron2017_bin10
405 prop_binCRTT_muon2017_bin15
406 prop_binCRWS_electron2018_bin11
407 prop_binSR_electron2017_bin0
408 prop_binCRTT_electron2016M_bin2
409 prop_binSR_electron2017_bin11_TVX
410 prop_binCRTT_electron2017_bin9
411 prop_binCRTT_muon2016M_bin11
412 prop_binCRTT_electron2018_bin4
413 prop_binCRTT_electron2018_bin11
414 prop_binCRTT_electron2016M_bin14
415 prop_binCRTT_electron2017_bin13
416 prop_binCRWS_electron2017_bin0
417 prop_binCRTT_muon2018_bin5
418 prop_binCRTT_muon2017_bin10
419 prop_binCRTT_muon2017_bin9
420 prop_binCRWS_electron2016M_bin15
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

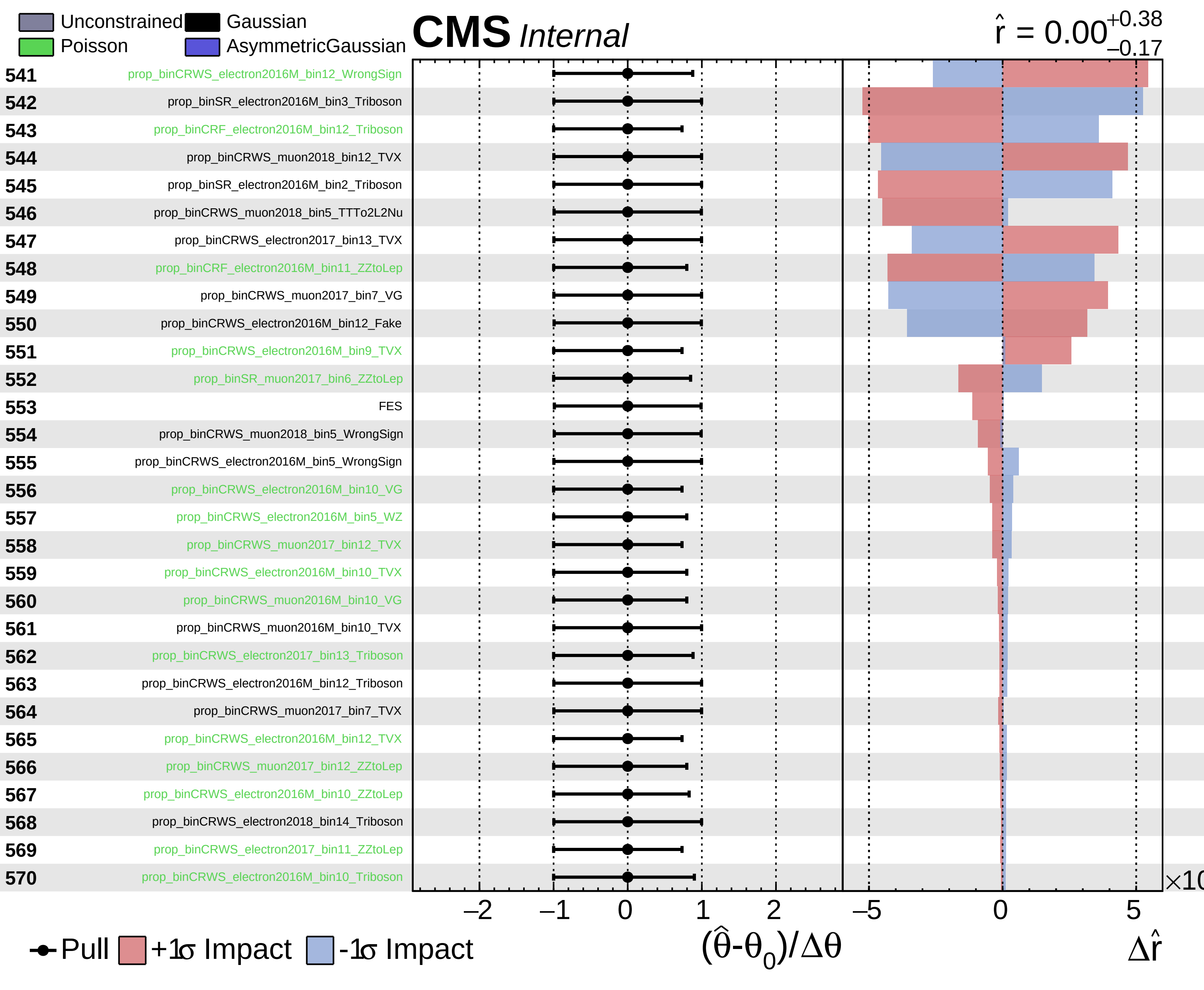

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
421 prop_binCRTT_muon2016M_bin8
422 prop_binCRTT_electron2017_bin4
423 lep_2017
424 prop_binCRTT_electron2018_bin3
425 prop_binCRF_electron2017_bin15_TVX
426 prop_binCRWS_muon2017_bin13
427 prop_binCRTT_muon2018_bin7
428 prop_binCRTT_muon2018_bin10
429 TES
430 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
431 prop_binCRTT_electron2017_bin7
432 prop_binCRTT_muon2017_bin16
433 prop_binCRF_electron2016M_bin12_VG
434 TTest_muon_2017
$1.00^{+0.04}_{-0.04}$
435 prop_binCRTT_electron2016M_bin4
436 prop_binCRWS_electron2016M_bin9_Fake
437 prop_binCRWS_muon2016M_bin15
438 lep_2018
439 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
440 prop_binCRTT_electron2018_bin12
441 prop_binCRTT_electron2018_bin5
442 prop_binCRTT_electron2016M_bin9
443 prop_binCRWS_electron2017_bin11_Fake
444 prop_binCRWS_muon2016M_bin14
445 prop_binCRF_electron2017_bin15_WrongSign
446 prop_binCRTT_electron2016M_bin12
447 tau_vsjet_2018
448 prop_binCRTT_electron2018_bin13
449 prop_binCRWS_muon2018_bin5_Fake
450 prop_binCRTT_muon2017_bin13
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



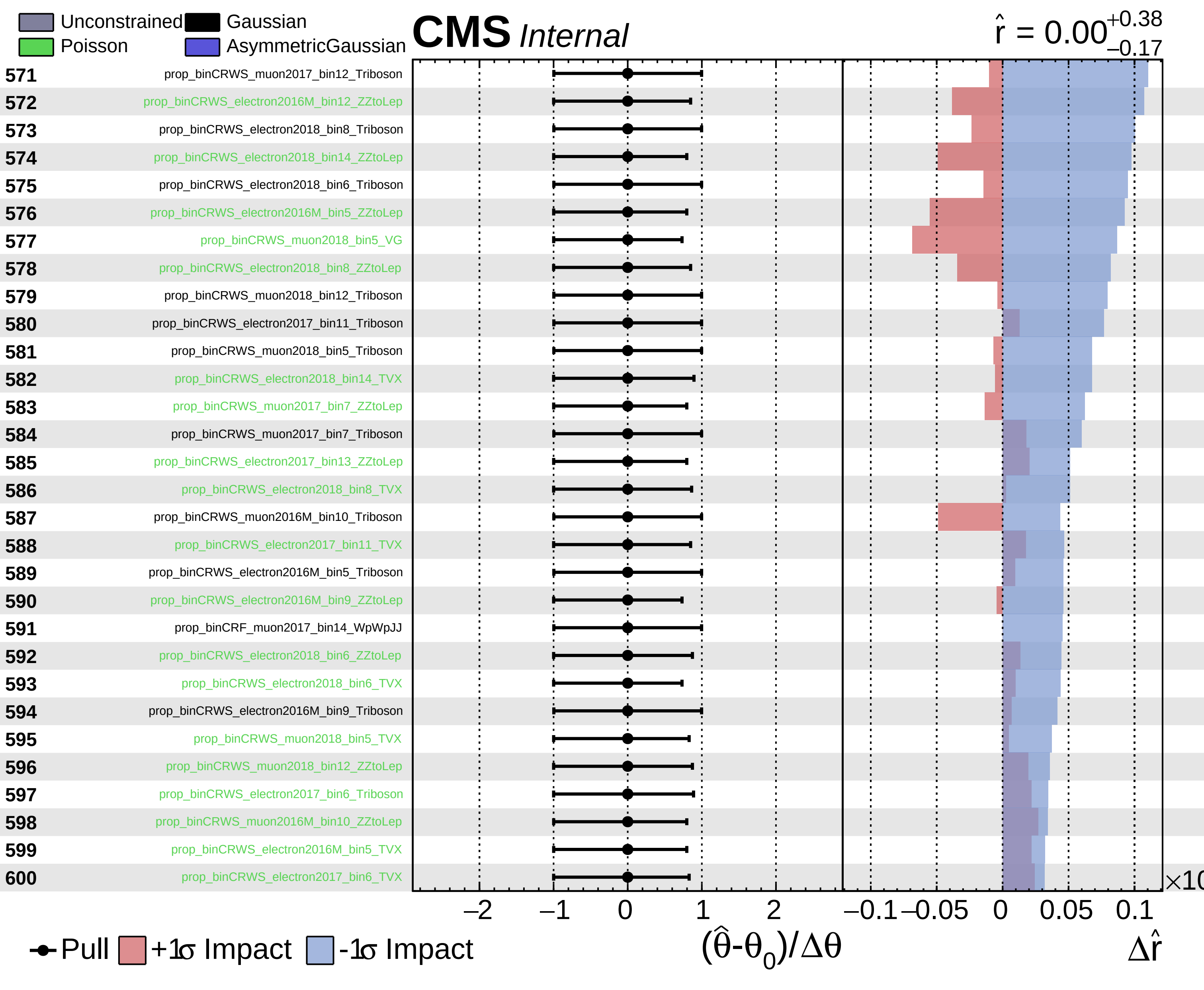
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
451 prop_binCRTT_electron2018_bin10
452 prop_binCRWS_muon2016M_bin10_Fake
453 prop_binCRTT_electron2017_bin15
454 prop_binCRF_electron2016M_bin12_Fake
455 prop_binCRTT_electron2017_bin5
456 prop_binCRTT_muon2018_bin14
457 prop_binCRTT_muon2017_bin7
458 prop_binCRTT_electron2016M_bin5
459 prop_binCRF_muon2016M_bin16
460 prop_binCRF_electron2016M_bin12_TTTo2L2Nu
461 prop_binCRTT_muon2016M_bin4
462 prop_binCRTT_muon2017_bin12
463 prop_binCRTT_electron2017_bin3
464 prop_binCRTT_muon2016M_bin5
465 prop_binCRTT_electron2018_bin15
466 prop_binCRTT_electron2016M_bin11
467 prop_binCRTT_muon2016M_bin2
468 prop_binCRTT_muon2017_bin6
469 prop_binSR_muon2017_bin6_WrongSign
470 prop_binCRTT_electron2016M_bin10
471 prop_binSR_electron2016M_bin2_WZ
472 prop_binSR_muon2017_bin6_TTTo2L2Nu
473 prop_binSR_muon2016M_bin0
474 prop_binCRWS_electron2017_bin13_Fake
475 prop_binCRTT_electron2018_bin6
476 prop_binCRTT_muon2017_bin8
477 prop_binSR_electron2016M_bin2_WrongSign
478 prop_binCRWS_muon2018_bin12_WrongSign
479 prop_binSR_electron2016M_bin3_VG
480 prop_binCRWS_electron2017_bin14
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
481 prop_binSR_electron2016M_bin2_TTTo2L2Nu
482 prop_binCRF_electron2017_bin14_Triboson
483 prop_binCRTT_muon2016M_bin14
484 prop_binCRWS_electron2018_bin14_WrongSign
485 prop_binCRTT_electron2016M_bin15
486 prop_binCRWS_electron2018_bin14_Fake
487 prop_binSR_electron2016M_bin3_TTTo2L2Nu
488 prop_binCRTT_electron2017_bin8
489 lep_2016M
490 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
491 prop_binCRWS_muon2017_bin12_Fake
492 prop_binCRWS_electron2016M_bin10_WrongSign
493 prop_binCRTT_muon2017_bin11
494 prop_binCRF_electron2016M_bin11_WrongSign
495 prop_binCRWS_muon2017_bin15
496 prop_binCRTT_electron2017_bin14
497 prop_binCRWS_muon2017_bin7_WrongSign
498 prop_binCRF_electron2016M_bin11_TVX
499 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
500 prop_binCRF_muon2017_bin14_WrongSign
501 prop_binCRF_electron2016M_bin12_ZZtoLep
502 prop_binCRF_muon2017_bin14_TVX
503 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
504 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
505 prop_binSR_electron2016M_bin3_ZZtoLep
506 prop_binSR_electron2016M_bin2_VG
507 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
508 prop_binCRF_electron2017_bin15_ZZtoLep
509 prop_binCRWS_muon2017_bin12_WrongSign
510 prop_binSR_electron2016M_bin2_ZZtoLep
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

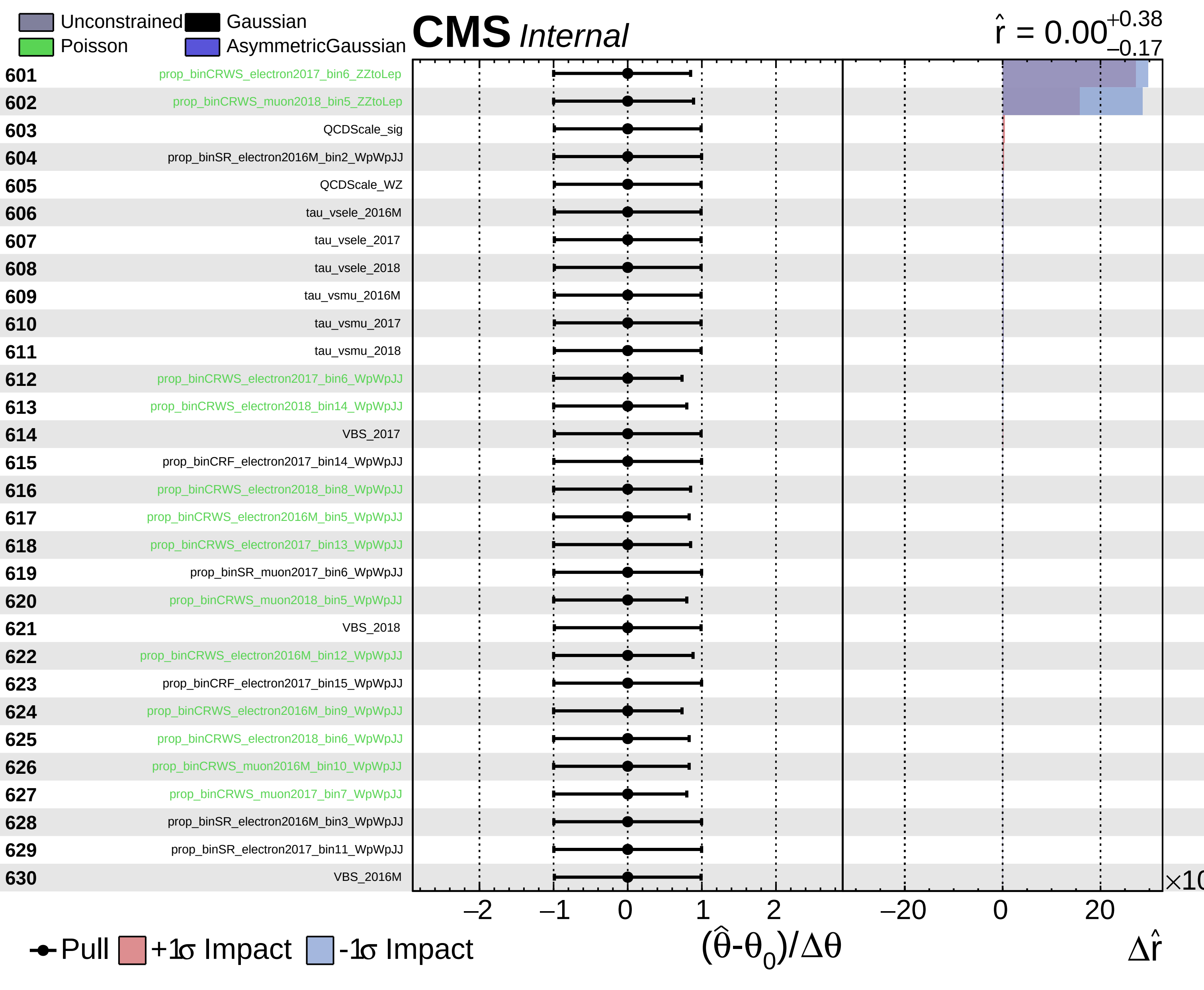

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.38 −0.17
511 prop_binCRWS_muon2016M_bin10_WrongSign
512 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
513 prop_binCRF_electron2016M_bin12_TVX
514 TTest_electron_2017
1.00 +0.05 −0.05
515 prop_binSR_electron2016M_bin3_WrongSign
516 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
517 prop_binCRF_electron2016M_bin11_Triboson
518 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
519 prop_binCRWS_electron2017_bin13_WrongSign
520 prop_binSR_muon2017_bin6_TVX
521 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
522 prop_binSR_electron2016M_bin3_TVX
523 prop_binCRWS_electron2018_bin6_WrongSign
524 prop_binCRWS_electron2017_bin11_WrongSign
525 prop_binCRF_muon2017_bin14_Triboson
526 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
527 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
528 prop_binSR_muon2017_bin6_Triboson
529 prop_binCRWS_electron2018_bin8_WrongSign
530 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
531 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
532 prop_binCRWS_electron2017_bin6_WZ
533 prop_binCRWS_electron2017_bin6_WrongSign
534 prop_binCRWS_electron2018_bin14_VG
535 prop_binCRWS_electron2017_bin13_WZ
536 prop_binCRWS_electron2016M_bin9_WrongSign
537 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
538 prop_binSR_electron2016M_bin2_TVX
539 prop_binCRWS_electron2018_bin8_VG
540 prop_binCRF_muon2017_bin14_ZZtoLep
×10
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



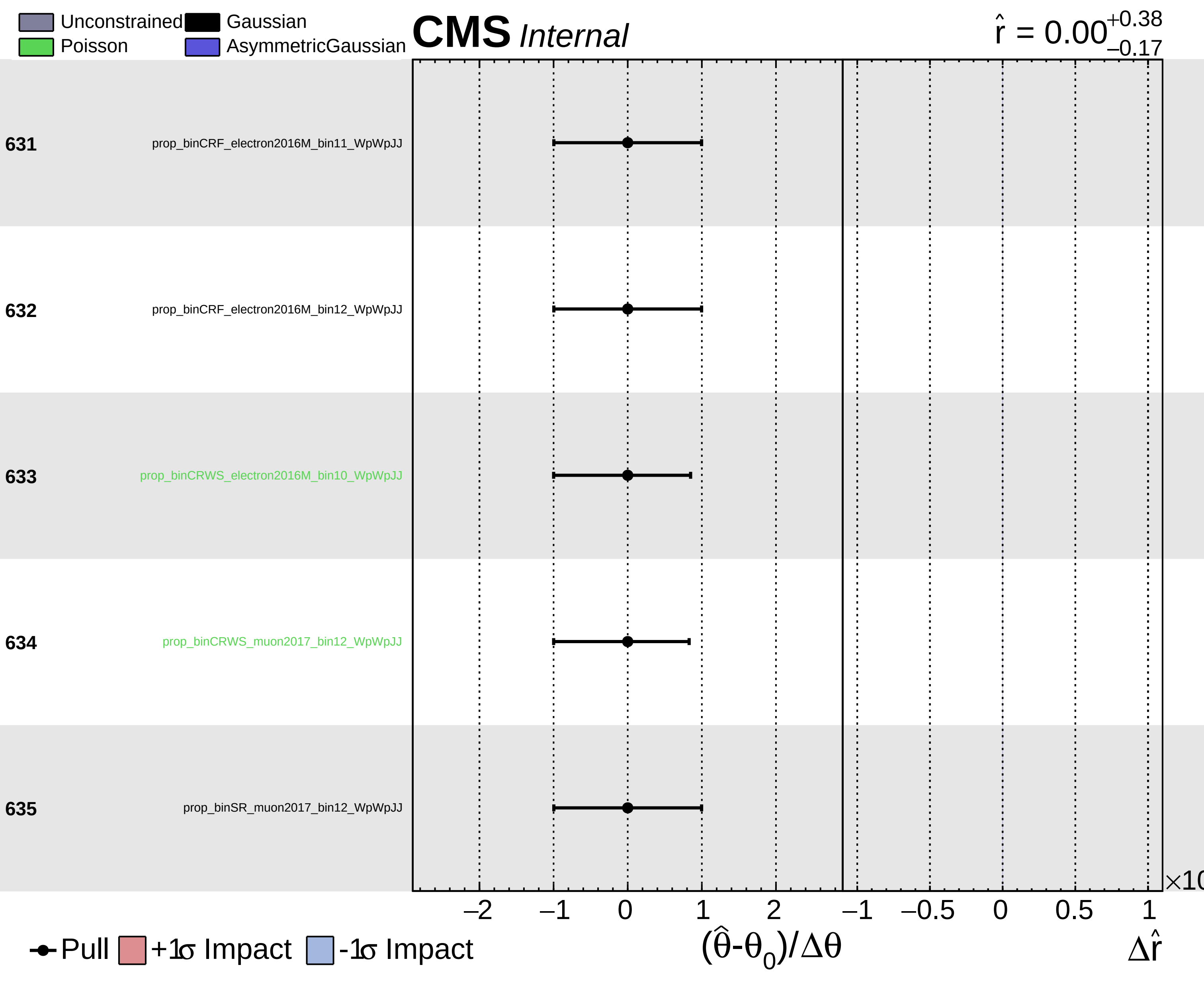
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
541 prop_binCRWS_electron2016M_bin12_WrongSign
542 prop_binSR_electron2016M_bin3_Triboson
543 prop_binCRF_electron2016M_bin12_Triboson
544 prop_binCRWS_muon2018_bin12_TVX
545 prop_binSR_electron2016M_bin2_Triboson
546 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
547 prop_binCRWS_electron2017_bin13_TVX
548 prop_binCRF_electron2016M_bin11_ZZtoLep
549 prop_binCRWS_muon2017_bin7_VG
550 prop_binCRWS_electron2016M_bin12_Fake
551 prop_binCRWS_electron2016M_bin9_TVX
552 prop_binSR_muon2017_bin6_ZZtoLep
553 FES
554 prop_binCRWS_muon2018_bin5_WrongSign
555 prop_binCRWS_electron2016M_bin5_WrongSign
556 prop_binCRWS_electron2016M_bin10_VG
557 prop_binCRWS_electron2016M_bin5_WZ
558 prop_binCRWS_muon2017_bin12_TVX
559 prop_binCRWS_electron2016M_bin10_TVX
560 prop_binCRWS_muon2016M_bin10_VG
561 prop_binCRWS_muon2016M_bin10_TVX
562 prop_binCRWS_electron2017_bin13_Triboson
563 prop_binCRWS_electron2016M_bin12_Triboson
564 prop_binCRWS_muon2017_bin7_TVX
565 prop_binCRWS_electron2016M_bin12_TVX
566 prop_binCRWS_muon2017_bin12_ZZtoLep
567 prop_binCRWS_electron2016M_bin10_ZZtoLep
568 prop_binCRWS_electron2018_bin14_Triboson
569 prop_binCRWS_electron2017_bin11_ZZtoLep
570 prop_binCRWS_electron2016M_bin10_Triboson
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
571 prop_binCRWS_muon2017_bin12_Triboson
572 prop_binCRWS_electron2016M_bin12_ZZtoLep
573 prop_binCRWS_electron2018_bin8_Triboson
574 prop_binCRWS_electron2018_bin14_ZZtoLep
575 prop_binCRWS_electron2018_bin6_Triboson
576 prop_binCRWS_electron2016M_bin5_ZZtoLep
577 prop_binCRWS_muon2018_bin5_VG
578 prop_binCRWS_electron2018_bin8_ZZtoLep
579 prop_binCRWS_muon2018_bin12_Triboson
580 prop_binCRWS_electron2017_bin11_Triboson
581 prop_binCRWS_muon2018_bin5_Triboson
582 prop_binCRWS_electron2018_bin14_TVX
583 prop_binCRWS_muon2017_bin7_ZZtoLep
584 prop_binCRWS_muon2017_bin7_Triboson
585 prop_binCRWS_electron2017_bin13_ZZtoLep
586 prop_binCRWS_electron2018_bin8_TVX
587 prop_binCRWS_muon2016M_bin10_Triboson
588 prop_binCRWS_electron2017_bin11_TVX
589 prop_binCRWS_electron2016M_bin5_Triboson
590 prop_binCRWS_electron2016M_bin9_ZZtoLep
591 prop_binCRF_muon2017_bin14_WpWpJJ
592 prop_binCRWS_electron2018_bin6_ZZtoLep
593 prop_binCRWS_electron2018_bin6_TVX
594 prop_binCRWS_electron2016M_bin9_Triboson
595 prop_binCRWS_muon2018_bin5_TVX
596 prop_binCRWS_muon2018_bin12_ZZtoLep
597 prop_binCRWS_electron2017_bin6_Triboson
598 prop_binCRWS_muon2016M_bin10_ZZtoLep
599 prop_binCRWS_electron2016M_bin5_TVX
600 prop_binCRWS_electron2017_bin6_TVX
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
601 prop_binCRWS_electron2017_bin6_ZZtoLep
602 prop_binCRWS_muon2018_bin5_ZZtoLep
603 QCDScale_sig
604 prop_binSR_electron2016M_bin2_WpWpJJ
605 QCDScale_WZ
606 tau_vsele_2016M
607 tau_vsele_2017
608 tau_vsele_2018
609 tau_vsmu_2016M
610 tau_vsmu_2017
611 tau_vsmu_2018
612 prop_binCRWS_electron2017_bin6_WpWpJJ
613 prop_binCRWS_electron2018_bin14_WpWpJJ
614 VBS_2017
615 prop_binCRF_electron2017_bin14_WpWpJJ
616 prop_binCRWS_electron2018_bin8_WpWpJJ
617 prop_binCRWS_electron2016M_bin5_WpWpJJ
618 prop_binCRWS_electron2017_bin13_WpWpJJ
619 prop_binSR_muon2017_bin6_WpWpJJ
620 prop_binCRWS_muon2018_bin5_WpWpJJ
621 VBS_2018
622 prop_binCRWS_electron2016M_bin12_WpWpJJ
623 prop_binCRF_electron2017_bin15_WpWpJJ
624 prop_binCRWS_electron2016M_bin9_WpWpJJ
625 prop_binCRWS_electron2018_bin6_WpWpJJ
626 prop_binCRWS_muon2016M_bin10_WpWpJJ
627 prop_binCRWS_muon2017_bin7_WpWpJJ
628 prop_binSR_electron2016M_bin3_WpWpJJ
629 prop_binSR_electron2017_bin11_WpWpJJ
630 VBS_2016M
−2 −1 0 1 2
−20 0 20
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.38}_{-0.17}$
631 prop_binCRF_electron2016M_bin11_WpWpJJ
632 prop_binCRF_electron2016M_bin12_WpWpJJ
633 prop_binCRWS_electron2016M_bin10_WpWpJJ
634 prop_binCRWS_muon2017_bin12_WpWpJJ
635 prop_binSR_muon2017_bin12_WpWpJJ
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$